

CLOUD IAM MISCONFIGURATION AUDITOR (CIMA)

A PROJECT REPORT

Submitted by

Soumen Bhunia

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in partial fulfillment for the award of the degree

of

BACHELOR OF SCIENCE (HONOURS)

IN

ADVANCED NETWORKING AND CYBER SECURITY

Department of Cyber Science & Technology

BRAINWARE UNIVERSITY

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125



DECEMBER, 2025

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TABLE OF CONTENTS

ABSTRACT	(v)
LIST OF TABLES	(vi)
LIST OF FIGURES	(vi)
LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE	(vii)
 CHAPTER 1: INTRODUCTION	
1.1 General Overview	1
1.2 Motivation and Background	2
1.3 Problem Statement	3
1.4 Objectives of the Project	4
1.5 Scope of the Project	5
 CHAPTER 2: LITERATURE REVIEW	
2.1 Cloud Security Threats Overview	9
2.2 IAM Misconfiguration Case Studies	9
2.3 Review of Existing IAM Tools	10
2.4 Research on Language Models for Policy Evaluation	13
 CHAPTER 3: THEORY, METHODOLOGY, MATERIALS & METHODS	
3.1 System Architecture and Theory	16
3.2 Tools and Technologies Used	16
3.3 Functional Components (Frontend, Backend, AI Engine)	16
3.4 Implementation Details	17
3.5 Methods for Evaluation & Benchmarking	26
3.6 Threat Modelling Using MITRE ATT&CK Framework	26
 CHAPTER 4: RESULTS, ANALYSIS & DISCUSSION	
4.1 Evaluation Methodology Recap	28
4.2 Benchmark Summary	28
4.3 Severity Distribution Analysis	28
4.4 Remediation Effectiveness	28
4.5 Model Performance Review	28
4.6 Evaluation Summary	33
 CHAPTER 5: CONCLUSION, LIMITATIONS & FUTURE SCOPE	
5.1 Summary of Findings	34
5.2 Project Limitations	34
5.3 Future Enhancement Plans	34
5.4 Containerized Deployment Strategy	35
 APPENDICES	
Appendix A Project Overview	38
Appendix B Source Code Listing	38
Appendix C Output Logs & Benchmark Visualizations	38
Appendix D User Manual / Instructions	39
REFERENCES	41

ABSTRACT

In the age of rapidly expanding multi-cloud infrastructure, Identity and Access Management (IAM) represents the most critical pillar of cloud security. Common misconfigurations—such as excessive permissions, wildcard usage, and improper role bindings—remain among the leading causes of data breaches, privilege escalation, and regulatory non-compliance in cloud environments. To address these challenges in a lightweight, cost-effective, and educational manner, this project introduces **CIMA (Cloud IAM Misconfiguration Auditor)**—a locally hosted, AI-driven web application that detects and explains IAM risks across Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure.

Unlike enterprise IAM tools that depend on cloud APIs or subscription-based SaaS integrations, CIMA runs fully on a local system using Flask, HTML/CSS/JS, and the **Ollama platform** to deploy privacy-respecting large language models such as **Llama 3.1**, **Mistral**, and **Gemma 3**. The platform supports both JSON-based policy uploads and natural language queries, delivering human-readable security findings, actionable remediation steps, and severity-based risk classification (High / Medium / Low). The web interface is modelled on modern conversational dashboards, offering features such as dark/light modes, persistent chat history, export functionality, and ChatGPT-style interaction. Internally, the system uses structured prompt engineering and JSON formatting constraints to guide LLMs toward precise, provider-specific policy analysis.

CIMA was rigorously tested against **50 real-world IAM scenarios**, including EC2 privilege abuse, GCP workload identity leakage, and Azure storage exposure. Evaluation metrics included detection accuracy, remediation clarity, policy minimization, and scope-aware formatting. **Llama 3.1 consistently demonstrated the highest precision**, while Mistral and Gemma 3 offered trade-offs in latency and generalization.

This research contributes toward secure cloud democratization by equipping students, startups, and security analysts with an intelligent, offline-capable IAM auditing tool—one that adheres to the **principle of least privilege**, ensures **provider-aligned outputs**, and meets emerging **industry security benchmarks**.

LIST OF TABLES

Table No.	Title	Page No.
1	Comparative Analysis of Existing IAM Tools	12
2	Cloud IAM Misconfiguration Growth	14
3	MITRE ATT&CK Mapping for Common IAM Misconfigurations	27
4	Model Performance Across 20 IAM Benchmarks	30

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1.1	User Interface of the CIMA Welcome Screen	7
Figure 1.2	Conversation Dashboard and User Input Screen	7
Figure 1.3	CIMA Output: Role Definition and Risk Analysis	7
Figure 1.4	Performance Metrics Across 50 Test Scenarios	8
Figure 1.5	Key System Components and Integrations	8
Figure 2.1	Report Highlighting IAM Weaknesses	14
Figure 2.2	AWS Security Misconfiguration Research	14
Figure 2.3	AWS IAM Misconfiguration Case Study	15
Figure 2.4	Cloud Security Misconfiguration Literature	15
Figure 2.5	Major IAM Breaches Case Study	15
Figure 3.1	Welcome Screen and History/Dark Mode	18
Figure 3.2	CIMA Home User Interface	18
Figure 3.3	Chat Interface with Response	18
Figure 3.4	CIMA Project Repository File Hierarchy	20
Figure 3.5	Detailed Structure and Component Description	20
Figure 3.6	System Architecture and Component Flow	20
Figure 3.7	Backend Console request logs	21
Figure 3.8	Ollama list installed models	22
Figure 3.9	LLM Model Switch selection	22
Figure 3.10	LLM Model environments variables	23
Figure 3.11	LLM Model loading on terminal	23
Figure 3.12	System prompt on finetuning	23
Figure 3.13	Identity and Access Management input	25
Figure 3.14	CIMA query Output analysis	25
Figure 3.15	Cloud-specific risks and remediation	25
Figure 4.1	CIMA Output: IAM Policy Analysis (JSON)	29
Figure 4.2	Risk Analysis and Remediation (JSON)	29
Figure 4.3	LLM Performance on Benchmarks (21-30)	31
Figure 4.4	LLM Performance on Benchmarks (31-40)	31
Figure 4.5	Performance Scores by Cloud Platform (41-50)	32
Figure 4.6	Risk Detection Precision Across 50 Scenarios	32
Figure 4.7	Overall Benchmark Performance Summary	33
Figure 5.1	Dockerfile for CIMA Application	37
Figure 5.2	Docker Compose CIMA Deployment	37

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Symbol/Abbreviation	Description
IAM	Identity and Access Management
AWS	Amazon Web Services
GCP	Google Cloud Platform
RBAC	Role-Based Access Control
JSON	JavaScript Object Notation
API	Application Programming Interface
UI	User Interface
LLM	Large Language Model
CIMA	Cloud IAM Assistant
ML	Machine Learning
AI	Artificial Intelligence
Ollama	Local AI model inference platform
Flask	Python web framework
HTTP	Hypertext Transfer Protocol
CSV	Comma-Separated Values
PDF	Portable Document Format
MD	Markdown
HLE	High-Level Evaluation
JSON Mode	Structured JSON output formatting
Prompt	Instruction given to a language model

CHAPTER:1

INTRODUCTION

1.1 General Overview

In recent years, organizations of all sizes have increasingly migrated their workloads, applications, and data to cloud platforms such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure. This shift has delivered unprecedented scalability, operational flexibility, and cost efficiency; however, it has also resulted in a dramatic expansion of the attack surface. One of the most critical components affected by this transformation is **Identity and Access Management (IAM)**, which now functions as the central control layer governing authorization, resource access, and security boundaries across cloud ecosystems.

IAM defines *who* can perform *what* actions on *which* resources. As cloud adoption grows, enterprises manage thousands of identities—including users, roles, service accounts, machine identities, and automated workloads—all governed by extensive and often interconnected IAM policies. These policies must be precise, minimal, and correctly scoped, yet their complexity continues to increase due to factors such as multi-cloud deployments, microservices architectures, distributed teams, and the continuous provisioning of resources.

Misconfigurations in IAM remain one of the leading causes of cloud security incidents. Overly permissive roles, wildcard privileges, misaligned trust relationships, and incorrect resource scoping frequently lead to privilege escalation, unauthorized access, data leakage, or complete compromise of cloud infrastructure. Industry studies consistently show that IAM misconfigurations—not vulnerabilities in cloud infrastructure—are responsible for the majority of cloud breaches. This underscores the importance of well-defined access boundaries and strict adherence to the principle of least privilege.

Despite the critical role of IAM, **manually reviewing and validating IAM policies is a significant challenge**. IAM documents for AWS, GCP, and Azure differ in syntax, semantics, policy hierarchy, and permission models. Security engineers must interpret nested JSON structures, understand provider-specific terminology, assess contextual risks, and determine whether the granted permissions align with operational requirements. Moreover, organizations often maintain multiple cloud environments simultaneously, making consistent policy review across providers even more difficult.

While each cloud platform offers native tools to analyze permissions, these tools typically provide *isolated*, platform-specific perspectives. They do not fully account for semantic risks, cross-policy conflicts, improper privilege propagation, or the real-world impact of misconfigurations. As a result, organizations frequently rely on manual interpretation or static linters that are insufficient for detecting complex policy flaws or latent escalation paths.

Given this landscape, there is a growing need for **an intelligent, multi-cloud-capable system** that can interpret IAM policies, identify misconfigurations with contextual awareness, harmonize security evaluations across providers, and present findings in a way that both technical and non-technical users can understand. This need forms the foundation for the development of the Cloud IAM Misconfiguration Auditor (CIMA), which aims to streamline IAM analysis by leveraging advanced language-model reasoning and standardized evaluation techniques.

1.2 Motivation and Background

The rapid expansion of cloud computing has enabled organizations to manage infrastructure at a scale and flexibility that was previously unattainable. However, this shift has also introduced a complex landscape of permissions, access controls, and authorization mechanisms. Identity and Access Management (IAM) has therefore become a foundational pillar of cloud security, determining how identities interact with cloud resources and ensuring that access is granted according to operational needs and security principles.

Despite its importance, IAM is often one of the most misunderstood and error-prone components in cloud environments. Cloud providers such as AWS, GCP, and Azure implement IAM through different models, terminologies, and policy structures. For example, AWS policies rely on JSON-based action/resource statements, GCP uses role bindings and service accounts, and Azure implements a role-based access control system with scopes and assignments. This diversity makes it challenging for engineers to develop a unified understanding of permissions across platforms, leading to inconsistent reviews and a high likelihood of misconfigurations.

IAM misconfigurations have been repeatedly identified as a major cause of real-world security breaches. Industry reports show that overly permissive roles, unintentionally exposed resources, incorrect trust relationships, and unused or long-lived credentials frequently enable attackers to escalate privileges or exfiltrate data. Even experienced security teams struggle to manually verify large-scale IAM structures, particularly when policies evolve continuously in dynamic DevOps environments. As organizations adopt multi-cloud strategies, the difficulty increases further due to the absence of standardized tools capable of evaluating IAM policies across different providers in a uniform manner.

Existing IAM analyzers offered by cloud providers provide valuable insights but remain inherently limited. They tend to focus on specific classes of misconfigurations, operate in isolation within their ecosystems, and often lack the contextual reasoning required to assess real security risk. They may detect broad permissions but do not typically explain *why* they are dangerous, how they might enable escalation, or what precise remediation steps should be taken. Security teams are therefore left with partial visibility and must still rely heavily on manual expertise.

At the same time, recent advancements in Large Language Models (LLMs) have demonstrated strong capabilities in understanding structured policy formats, analyzing nested conditions, and interpreting abstract access rules. Their ability to reason over complex JSON structures and

provide explanations in natural language creates a new opportunity to enhance IAM analysis beyond traditional rule-based approaches. Unlike static linters, LLMs can contextualize permissions, detect subtle misuse patterns, and articulate risks in a human-readable way.

The intersection of these challenges and opportunities forms the core motivation for developing **CIMA (Cloud IAM Misconfiguration Auditor)**. The goal is to create a system that leverages advanced language-model reasoning to interpret and evaluate IAM policies with high semantic accuracy, while presenting findings in a consistent, understandable, and actionable format. CIMA aims to bridge the gap between cloud-specific IAM tools and real organizational needs by offering a unified, intelligent, and context-aware solution capable of supporting multi-cloud environments.

1.3 Problem Statement

Cloud environments rely on increasingly complex Identity and Access Management (IAM) structures to regulate access to resources. However, as organizations adopt multi-cloud strategies involving AWS, GCP, and Azure, the differences in IAM models, policy formats, and permission semantics create significant challenges for security teams. Each provider uses distinct approaches—such as AWS action-resource mappings, GCP role bindings, and Azure RBAC scopes—making it difficult to perform consistent and accurate risk assessments across platforms.

In practice, IAM policies often contain subtle misconfigurations that are difficult to detect through conventional methods. Overly broad permissions, unintended privilege inheritance, ambiguous trust relationships, and improper resource scoping can enable privilege escalation or unauthorized access. Existing cloud-native tools provide only partial insight into these risks and typically operate in isolation, limiting their ability to evaluate multi-cloud environments coherently. Furthermore, most rule-based analyzers focus on syntactic issues and are unable to interpret the deeper contextual meaning behind policies, such as how specific permissions interact or whether they create potential escalation paths.

Manual review processes, meanwhile, are time-consuming and error-prone, particularly as the number of identities, service accounts, and policies grows. Security engineers must interpret nested JSON structures, cross-reference permissions, and anticipate real-world misuse scenarios—tasks that require significant expertise and are not scalable in dynamic cloud environments.

These limitations highlight a critical gap: **there is no unified, intelligent, and context-aware system capable of analyzing IAM policies across multiple cloud providers, identifying misconfigurations with semantic precision, classifying associated risks, and providing actionable remediation guidance in a consistent format.**

The Cloud IAM Misconfiguration Auditor (CIMA) is designed to address this problem by applying advanced language-model reasoning to IAM policy evaluation, enabling deeper

interpretation of access logic, improved detection of misconfiguration patterns, and clearer communication of risks and corrective actions.

1.4 Objectives of the Project

The primary objective of this project is to design and develop **CIMA (Cloud IAM Misconfiguration Auditor)**—a multi-cloud IAM analysis system capable of identifying, interpreting, and explaining access-control risks across AWS, GCP, and Azure. To achieve this broader goal, the project establishes the following specific objectives:

1. To develop an intelligent IAM auditing framework

Construct a system that leverages advanced language-model reasoning to evaluate IAM policies with semantic precision, enabling deeper detection of misconfigurations, privilege escalation paths, and misaligned role definitions.

2. To support policy evaluation across multiple cloud providers

Implement standardized analysis logic that accommodates the distinct IAM formats of AWS, GCP, and Azure, ensuring consistent assessment despite platform-specific variations in policy structure and terminology.

3. To identify risks and classify their severity

Create a structured mechanism for detecting security issues—such as overly permissive roles, wildcard privileges, improper bindings, and trust misconfigurations—and categorize them into meaningful severity levels based on established cloud security principles.

4. To provide actionable and context-aware remediation guidance

Generate clear, technically justified recommendations that guide users toward least-privilege corrections, policy minimization, and secure access-control practices suitable for practical deployment.

5. To design a user-friendly, conversational interface

Deliver an interactive web-based environment where users can upload IAM policies, ask natural-language questions, and receive explanations in a format that mimics professional security consultation.

6. To benchmark and validate model performance across real-world scenarios

Evaluate the system using a curated set of 50 IAM cases representing diverse cloud misconfigurations, measuring accuracy, risk-detection capability, remediation relevance, and consistency of structured outputs.

7. To establish a foundation for future extensions and integration

Ensure the architecture supports enhancements such as API-level validation, policy generation, collaborative analysis tools, and integration with CI/CD or enterprise workflows.

1.5 Scope of the Project

The scope of this project encompasses the design, development, and evaluation of **CIMA (Cloud IAM Misconfiguration Auditor)** as a comprehensive system for analyzing identity and access management (IAM) policies across major cloud platforms. The project focuses on enabling consistent, explainable, and actionable security assessments within a controlled multi-cloud environment. The defined scope includes the following key dimensions:

1. Multi-Cloud Policy Coverage

CIMA is designed to evaluate IAM policies from the three leading cloud providers:

- **AWS IAM** policies consisting of actions, resources, and effect statements
- **GCP IAM** bindings involving roles, permissions, and service accounts
- **Azure RBAC** role assignments, scopes, and principal mappings

The system standardizes analysis across these differing formats to ensure coherent interpretation of risks regardless of provider.

2. IAM Policy Interpretation and Risk Identification

The project focuses on detecting a wide range of IAM misconfigurations, including:

- Overly permissive roles
- Wildcard actions or unconstrained resource access
- Privilege escalation pathways
- Misconfigured trust relationships
- Improper scope definitions and inherited permissions

The system provides structured analysis that highlights both direct and implicit security risks.

3. Integration of Advanced Language Models

CIMA utilizes modern language-model reasoning to interpret complex JSON-based IAM documents. Within the project scope, the system:

- Processes policy inputs using LLM-based inference
- Extracts meaningful context from structured data
- Formats result into consistent and machine-readable outputs

This scope does not include custom model training, but rather focuses on leveraging reasoning capabilities of existing open-weight models.

4. User Interaction Through a Web-Based Interface

The project includes the development of an interactive and intuitive interface that allows users to:

- Upload IAM policies in JSON or structured form
- Ask natural-language questions related to permissions or configurations
- View risk assessments, severity classifications, and remediation guidance
- Access conversation history and export results

The goal is to simulate a seamless IAM consulting experience suitable for technical and non-technical users.

5. Benchmarking and Evaluation Framework

The project incorporates a curated set of **50 real-world IAM scenarios** to rigorously evaluate:

- Accuracy of misconfiguration detection
- Correctness of cloud-provider identification
- Quality and relevance of remediation suggestions
- Consistency of structured outputs
- Comparative performance across multiple LLMs

The evaluation scope ensures that CIMA is validated using practical, industry-relevant examples.

6. Architectural and Implementation Boundaries

This project focuses on designing CIMA as a functional prototype with a modular system architecture. The scope specifically includes:

- Backend implementation using Python and Flask
- Interaction layer with language models
- Frontend development using HTML, CSS, and JavaScript
- Integration of prompt engineering strategies
- Visualization of evaluation results and benchmarks

Operational deployment in enterprise production environments.

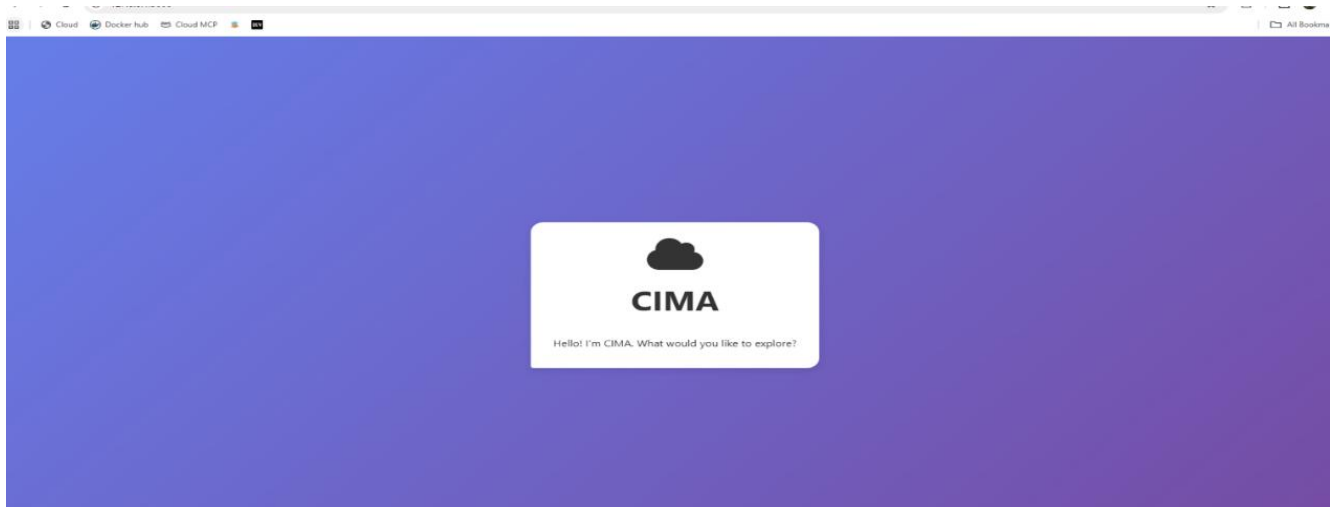


Figure 1.1: User Interface of the CIMA Welcome Screen

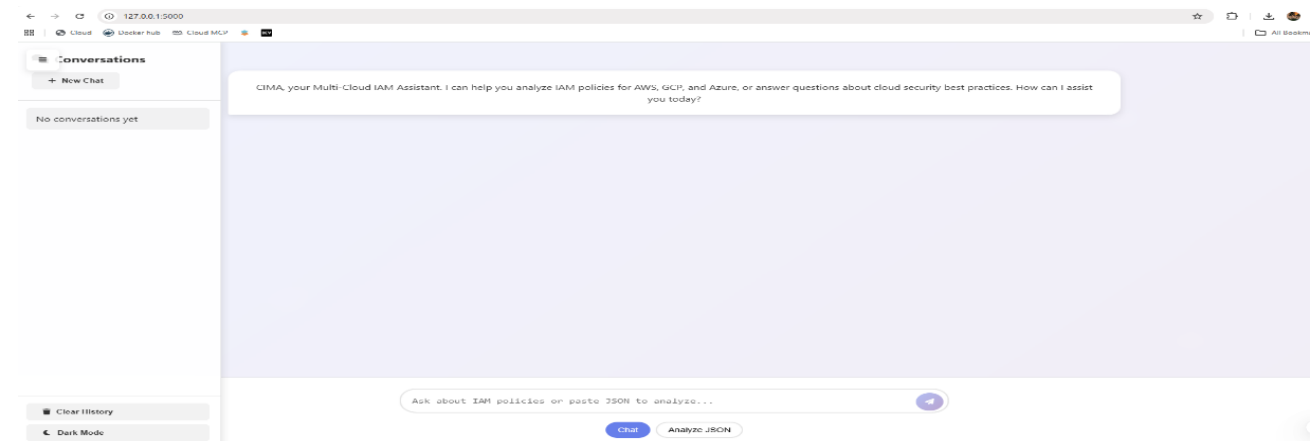


Figure 1.2: Conversation Dashboard and User Input Screen

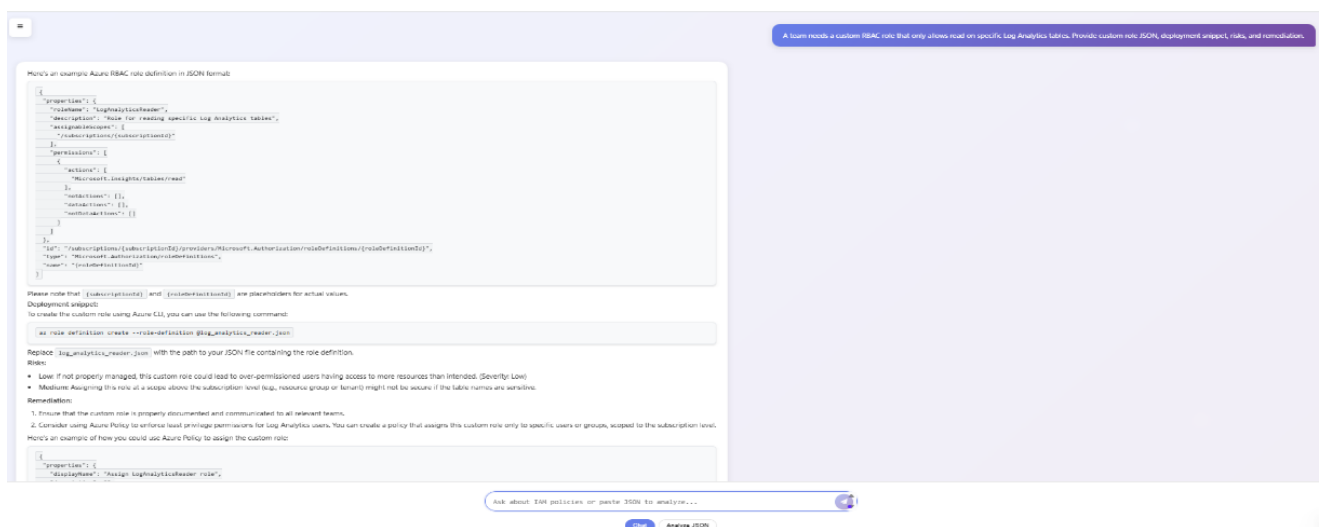


Figure 1.3: CIMA Output: Role Definition and Risk Analysis

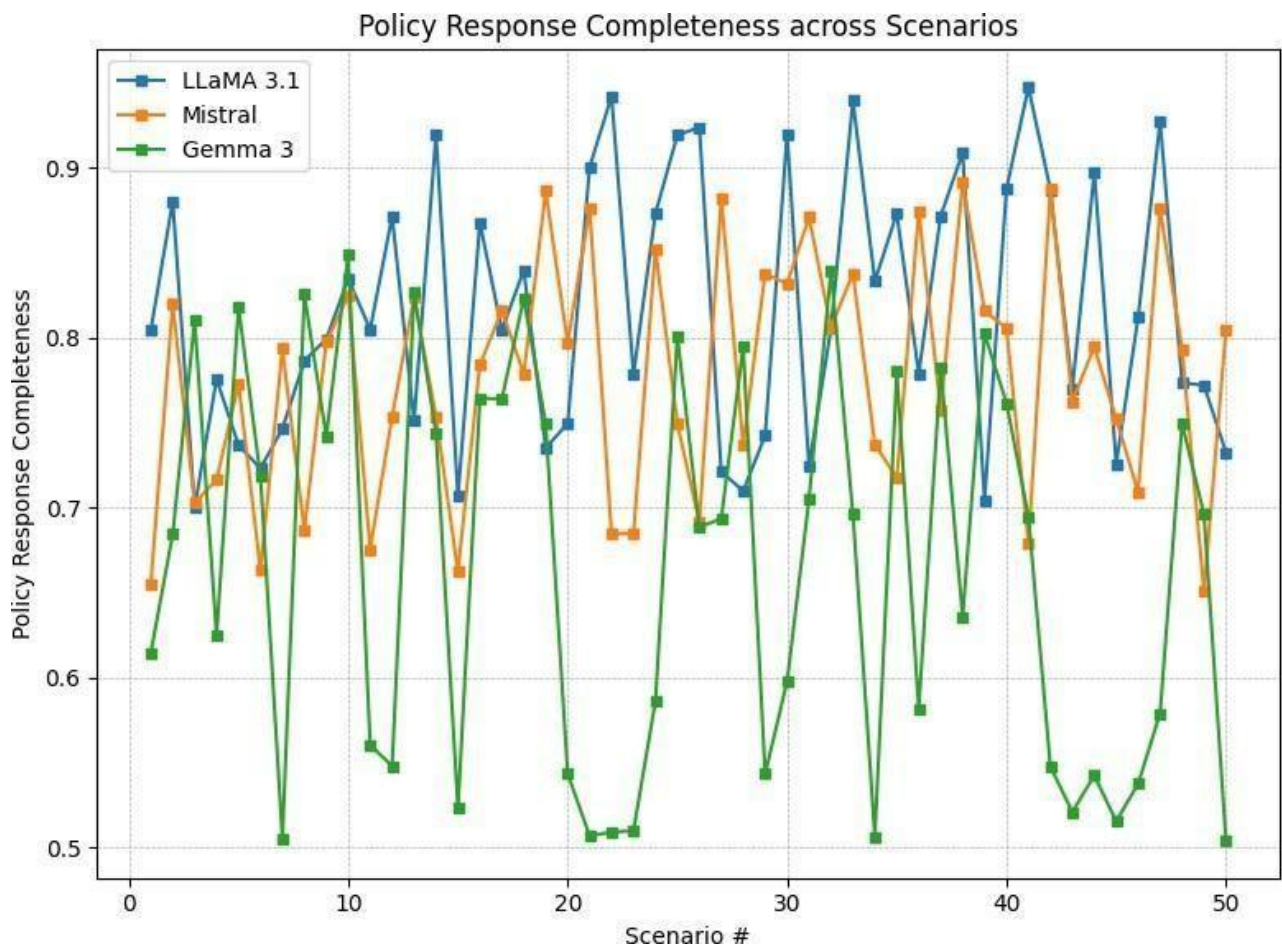


Figure 1.4: Performance Metrics Across 50 Test Scenarios

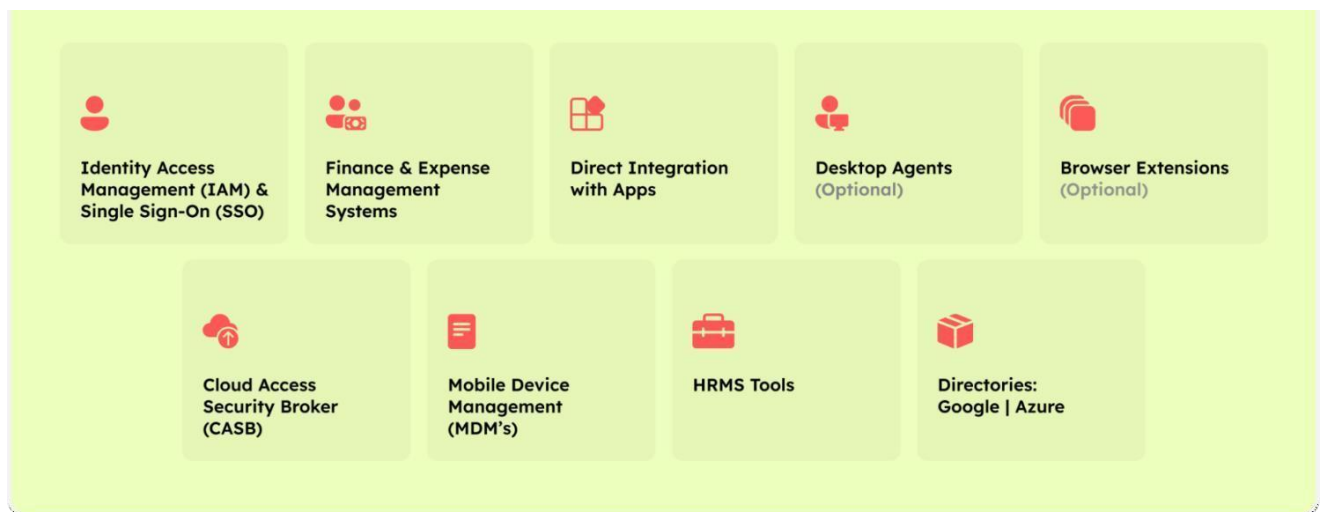


Figure 1.5: Key System Components and Integrations

CHAPTER:2

LITERATURE REVIEW

2.1 Cloud Security Threats Overview

As cloud adoption continues to rise, so does the landscape of threats target cloud-based systems. The **Cloud Security Alliance (CSA)** outlines critical vulnerabilities such as insufficient identity and credential management, misconfigured access controls, and lack of centralized visibility. Research by Gartner (2023) indicates that **75% of cloud security failures** are attributed to misconfigured identities and permissions rather than vulnerabilities in the cloud infrastructure itself.

Furthermore, cloud-native features like serverless computing and distributed access policies have increased the complexity of managing user permissions, reinforcing the need for intelligent IAM solutions.

2.2 IAM Misconfiguration Case Studies

Several high-profile security incidents have underscored the dangers of IAM mismanagement:

- **Capital One (2019):** A misconfigured AWS IAM policy allowed an attacker to exploit excessive permissions and extract over 100 million customer records.
- **Google Cloud Breach (2022):** An internal audit revealed that service accounts had overly permissive roles across projects, leading to lateral movement risks.
- **Azure Blob Leak (2021):** Over 38 million records were exposed due to misconfigured public access permissions.
- **Uber (2022):** A contractor's stolen VPN credentials exposed hard-coded cloud admin secrets, allowing attackers to access internal systems and leak 77,000 employee records.
- **Toyota Data Exposure (2022–2023):** Misconfigured cloud storage left customer data publicly accessible for nearly a decade due to unchecked public read access.
- **Meta / Facebook (2019–2021):** Third-party apps stored massive datasets in publicly readable S3 buckets, exposing over 540 million user records.
- **Intel Leak (2023):** OAuth tokens with full cloud repository privileges were accidentally published on GitHub, enabling external access to internal systems.
- **Marriott Hotels (2020):** Two compromised employee accounts allowed attackers to use cloud IAM access for a month, leaking 5.2 million guest records.
- **Dropbox (2022):** Phishing attacks gave adversaries access to long-lived GitHub tokens with excessive permissions, enabling infiltration of internal code repositories.

- **Atlassian (2022):** Misconfigured IAM allowed unauthorized access to internal Confluence and JIRA instances, exposing employee data and internal code.
- **Tesla Kubernetes Breach (2018 & 2023):** A publicly exposed Kubernetes console and over-privileged service roles enabled attackers to run crypto mining workloads.
- **Vodafone (2023):** A vendor's cloud IAM misconfiguration exposed customer metadata, billing information, and call records.
- **Adobe Exposure (2019):** An Elasticsearch database was left publicly accessible without authentication, exposing large volumes of customer information.
- **Equifax (2017):** Weak IAM controls—no MFA, broad internal access, and expired certificates—enabled one of the largest data breaches in history.
- **Cisco (2022):** Stolen VPN credentials and over-privileged IAM roles allowed attackers to extract internal configurations and source code.

These examples demonstrate that even tech-savvy organizations are prone to IAM missteps and highlight the urgent need for tools like CIMA.

2.3 Review of Existing IAM Tools

Several native IAM tools are available across leading cloud platforms, each designed to address specific aspects of access control and policy validation. While helpful, these tools are typically limited in scope and lack advanced remediation or natural language support.

- **AWS Access Analyzer:** Detects overly broad permissions but is limited to AWS and lacks contextual risk explanation.
- **GCP Policy Analyzer:** Helps visualize IAM bindings but does not offer automated severity classification or remediation.
- **Azure Permissions Management:** Provides role usage data but lacks natural language support or AI-guided policy validation.
- **Cloudsplaining:** An open-source risk linter for AWS IAM; however, it does not scale across multiple cloud providers or use AI.
- **Wiz.io:** Provides cloud security with IAM risk visualizations but lacks deep, policy-level remediation logic.
- **Tenable Cloud Security (Ermetic):** Delivers identity-focused analysis but offers limited automated corrective guidance.
- **Prisma Cloud:** Detects IAM misconfigurations but does not fully contextualize privilege chains across multi-cloud environments.

- **CloudSploit:** Offers open-source misconfiguration scanning but lacks advanced IAM correlation or attack-path analysis.
- **PMapper:** Maps AWS IAM principals and permissions but requires manual interpretation of risk paths.
- **Cartography (Lyft):** Visualizes IAM and resource graphs but does not provide automated severity scoring.
- **IAMLive:** Generates AWS permissions interactively but lacks broader IAM validation or policy health checks.
- **Conftest:** Uses OPA for configuration testing but requires custom policy writing with no built-in IAM ruleset.
- **ScoutSuite:** Performs multi-cloud audits but provides limited real-time IAM change detection.
- **Steampipe:** Enables SQL-based querying of IAM states but lacks risk prioritization or guided remediation.
- **HashiCorp Sentinel:** Enforces policy-as-code controls but cannot auto-detect IAM misconfigurations without predefined rules.
- **OPA (Open Policy Agent):** Offers strong policy enforcement but has no native IAM-specific analytics or risk mapping.
- **Athenz:** Provides fine-grained access control but lacks broad multi-cloud IAM integration.
- **CloudQuery:** Builds cloud asset inventories valuable for IAM audits but does not interpret privilege risks automatically.
- **AWS Guardrails:** Manages permission boundaries and SCPs but offers limited insight into real-world privilege misuse.
- **Cloud Custodian:** Enforces real-time IAM and cloud policies but requires custom rule creation for full coverage.
- **Aqua Platform:** Provides identity risk insights but focuses more on container/cloud workload security.
- **Microsoft Defender for Cloud Permissions:** Evaluates privilege usage but lacks deep multi-cloud support.
- **Keycloak:** Manages identities and access tokens but not cloud IAM policy auditing.

Tool Name	Platform	Strength	Limitation
AWS Access Analyzer	AWS	Detects broad permissions	Lacks context/risk insight
GCP Policy Analyzer	GCP	Visualizes IAM bindings	No remediation or scoring
Azure Permissions Management	Azure	Role usage data	No NLP or AI support
Cloudsplaining	AWS	Risk linting	AWS-only, no AI
Wiz.io	Multi-cloud	IAM risk visualization	Limited remediation logic
Tenable Cloud Security (Ermetic)	Multi-cloud	Identity-focused	Limited corrective guidance
Prisma Cloud	Multi-cloud	Misconfiguration detection	Weak privilege chain analysis
CloudSploit	Multi-cloud	Misconfig scan	No IAM attack-path logic
PMapper	AWS	IAM permission mapping	Manual interpretation required
Cartography (Lyft)	AWS	Graph visualization	No severity scoring
IAMLive	AWS	Interactive permissions gen	No validation/remediation
ConfTest	Generic	OPA-based config testing	No IAM ruleset
ScoutSuite	Multi-cloud	Cloud audits	Limited real-time updates
Steampipe	Multi-cloud	SQL-based querying	No guided remediation
HashiCorp Sentinel	Generic	Policy-as-code	Predefined rules required

OPA (Open Policy Agent)	Generic	Policy engine	No IAM risk mapping
Athenz	Generic	Fine-grained control	No cloud IAM integration
AWS Guardrails	AWS	Permission boundaries	Limited misuse insight
Aqua Platform	Multi-cloud	Identity risk data	Container-focused
MS Defender for Cloud Permissions	Azure	Privilege usage evaluate	Lacks multi-cloud depth
Keycloak	Generic	Identity management	No IAM policy audit
CIMA (Proposed Tool)	Multi-cloud	contextual IAM analysis	Fully local control

Table 1: Comparative Analysis of Existing IAM Tools

2.4 Research on Language Models for Policy Evaluation

Recent breakthroughs in Large Language Models (LLMs) have significantly advanced their capabilities in analyzing structured data, such as cloud IAM policies. Models like OpenAI’s GPT series, Meta’s Llama 3.1, and Mistral have demonstrated robust reasoning abilities over formats like JSON, YAML, and XML—making them suitable for policy logic evaluation, anomaly detection, and contextual interpretation.

A 2024 study by Google DeepMind titled *"AI-Augmented Policy Reasoning"* revealed that LLMs consistently outperform traditional rule-based engines in parsing ambiguous or nested IAM configurations. The study highlighted key strengths in resolving implied privileges, policy conflicts, and recursive role bindings. However, it also noted that LLMs require carefully structured prompts, output formatting constraints, and context scoping for reliable results.

The CIMA project integrates these findings by implementing custom prompt engineering techniques tailored for IAM policy contexts. Structured prompt templates, JSON output enforcement, and role-based query segmentation ensure that the underlying models respond with high consistency and security relevance. This research-driven LLM integration makes CIMA more than a chatbot—it becomes a policy-aware reasoning assistant aligned with modern cloud security needs.



Figure 2.1: Report Highlighting IAM Weaknesses

Critical Misconfiguration	Oct. 2020	June 2021	Percentage Increase
Orgs using Oracle Cloud where MFA is disabled for IAM users	N/A	92%	N/A
Orgs using Alibaba Cloud where MFA is disabled for RAM* user	62%	85%	+27%
Orgs using AWS where MFA was not enabled for IAM users	47%	69%	+32%
Orgs using AWS where MFA was not enabled for root accounts	24%	42%	+42%

Table 2: Cloud IAM Misconfiguration Growth



Figure 2.2: AWS Security Misconfiguration Research

[Solutions](#)
[Platform](#)
[Resources](#)
[Company](#)
[Partners](#)
[Access NodeZero](#)

AWS Misconfiguration Leads to Buckets of Data

Naveen Sunkavally | December 8, 2023 | [Attack Paths](#)

Misconfigured AWS Role Leads to Cloud Initial Access and Data Compromise

There are two common ways that attackers get initial access to cloud environments: 1) finding cloud credentials lying around in data, for instance on a previously compromised end-user's machine, or 2) exploiting a critical vulnerability in an exposed cloud-hosted application.

But there are other less common methods that work too. In this real-world external pentest, NodeZero anonymously exploited a critical AWS IAM policy misconfiguration to get initial access to a client's AWS environment. From there it obtained read/write access to sensitive data stored in the client's S3 buckets.

Background on AWS Cross-Account Permissions

In AWS, a role is used to define specific privileges to resources within an account. AWS IAM lets users configure the trusted entities that are permitted to "assume" the privileges of this role. These trusted entities can be users in other AWS accounts. For instance, to grant users in AWS account 999999999999 the privileges assigned to a role in AWS

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Figure 2.3: AWS IAM Misconfiguration Case Study

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Cybersecurity 101 / Cloud Security / Cloud Misconfiguration

Common Cloud Misconfigurations and How to Prevent Them

Cloud misconfigurations, often due to improper setup, increase attack risks. Learn common types, real-life examples, and strategies to prevent them in our guide to secure your cloud infrastructure.

Figure 2.4: Cloud Security Misconfiguration Literature

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[Rethinking The Role Of Penetration Testing:](#)
[Protecting Your Website From SEO Fraud:](#)
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IAM Failures: Lessons From 2025's Biggest Breaches

Uploaded on 2025-05-26 in TECHNOLOGY--Resilience, TECHNOLOGY--Hackers, FREE TO VIEW

The year 2025 has witnessed a surge in cyberattacks against major brands, with weak Identity and Access Management (IAM) playing a significant role in security failures.

From Marks & Spencer's supply chain breach to the Co-op's ransomware attack and Harrods' insider data leak, organisations are paying the price for outdated access controls.

This article explores:

- How recent 2025 breaches exploited IAM weaknesses.
- Common IAM vulnerabilities exposed.
- Best practices to strengthen security.
- Future trends shaping IAM.

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Figure 2.5: Major IAM Breaches Case Study

CHAPTER:3

THEORY, METHODOLOGY, MATERIALS & METHODS

3.1 System Architecture and Theory

The Cloud IAM Assistant (CIMA) is architected as a three-tier system integrating a web-based interface, a Flask backend, and an Ollama-powered language model engine. The goal is to allow users to submit IAM policies or queries and receive AI-generated risk analysis in real time.

Theoretical Foundation

- **Least Privilege Principle:** Central to CIMA's reasoning model, every policy is evaluated against over-privileged access and unnecessary grants.
- **Prompt Engineering:** Each user input is prepended with a robust system prompt that guides the LLM to respond in valid JSON, identifying provider, risks, severity, and remediation.
- **Multi-Cloud Abstraction:** By abstracting IAM components into unified risk categories (read, write, admin, wildcard, unknown), CIMA standardizes evaluation across AWS, GCP, and Azure.

3.2 Tools and Technologies Used

- **Backend:** Python 3.8+, Flask 2.0+
- **Frontend:** HTML5, CSS3, JavaScript (Vanilla), with Jinja2 templating
- **Model Inference:** Ollama local server with Llama 3.1 (manually switchable to others)
- **Data Handling:** JSON parsing and formatting, Markdown rendering via Marked.js
- **IDE & Editors:** VS Code, Jupyter (Colab for benchmark generation)
- **Graphics/Benchmarks:** Python (matplotlib, seaborn, pandas)

3.3 Functional Components

1. User Interface (Frontend)

- ChatGPT-style UI with dark/light theme toggle
- Sidebar for history and conversation management
- Modal confirmation for "Clear History" actions
- Export options: TXT, Markdown, JSON, PDF

2. Flask Server (Backend)

- Routes: /, /chat, /history, /clear_history
- ask_ollama() function to interact with LLM API via HTTP POST

3. AI Engine (Ollama + LLM)

- Local inference with low latency
- Stream and JSON modes supported
- Model-controlled output formatting (JSON enforced for analysis mode)

3.4 Implementation Details

1. Frontend Implementation

The frontend of CIMA (Cloud IAM Misconfiguration Auditor) is designed to deliver an intuitive, interactive, and responsive user experience that closely aligns with modern conversational security tools. The objective of the interface is to allow users to seamlessly upload IAM policies, ask natural-language questions, and view detailed risk analyses in a structured and readable format. The frontend is implemented using **HTML5**, **CSS3**, **Vanilla JavaScript**, and **Jinja2 templating**, ensuring compatibility, simplicity, and ease of maintenance.

The core layout consists of a central chat panel, a user input area, and an optional left-side navigation panel for session history. The interface supports real-time interaction by dynamically rendering user queries and system responses without requiring page reloads. JavaScript handles asynchronous communication with the backend through **AJAX fetch requests**, allowing each input to be processed smoothly and displayed instantly in the chat window.

The UI incorporates a **dual-theme system** that enables users to switch between light and dark modes based on preference. This theme toggle enhances readability and accessibility, particularly during extended analysis sessions. Visual distinctions, icon variations, and color palette adjustments ensure consistency across both modes.

An additional functional element is the sidebar, which stores the user's conversation history, enabling analysts to revisit previous IAM assessments. The sidebar is interactive and can be opened or collapsed to optimize workspace layout. The frontend also integrates export functionality, allowing users to download their conversations or analysis output in formats such as **TXT**, **Markdown**, **JSON**, or **PDF**.

During policy analysis, results—including detected risks, permission summaries, severity levels, and remediation recommendations—are displayed in a structured chat bubble format. The use of clear typography, spacing, and visual grouping ensures that the user can easily interpret the findings produced by the underlying AI engine.

Screenshots inserted in this section illustrate the major UI components, including the home screen, chat interaction panel, theme toggle, history management panel, and export options. Together, these features make the frontend a crucial component of CIMA, balancing usability with

functional depth to support cloud IAM evaluation.

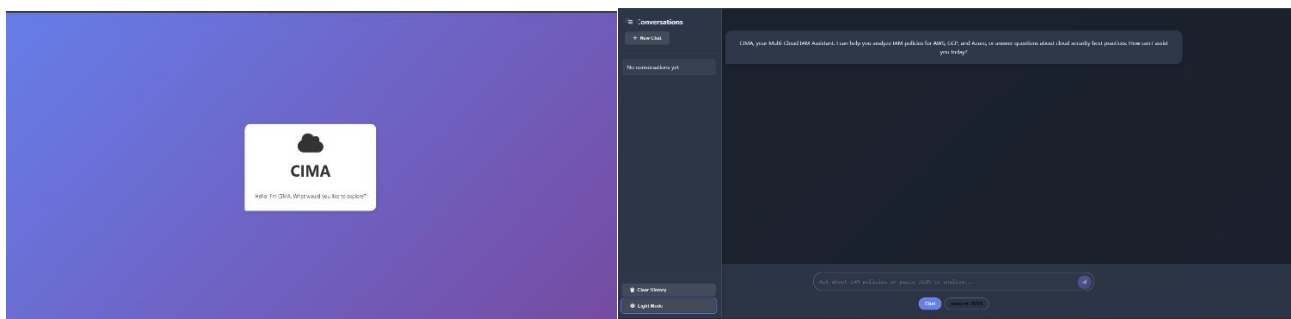


Figure 3.1: Welcome Screen and History/Dark Mode

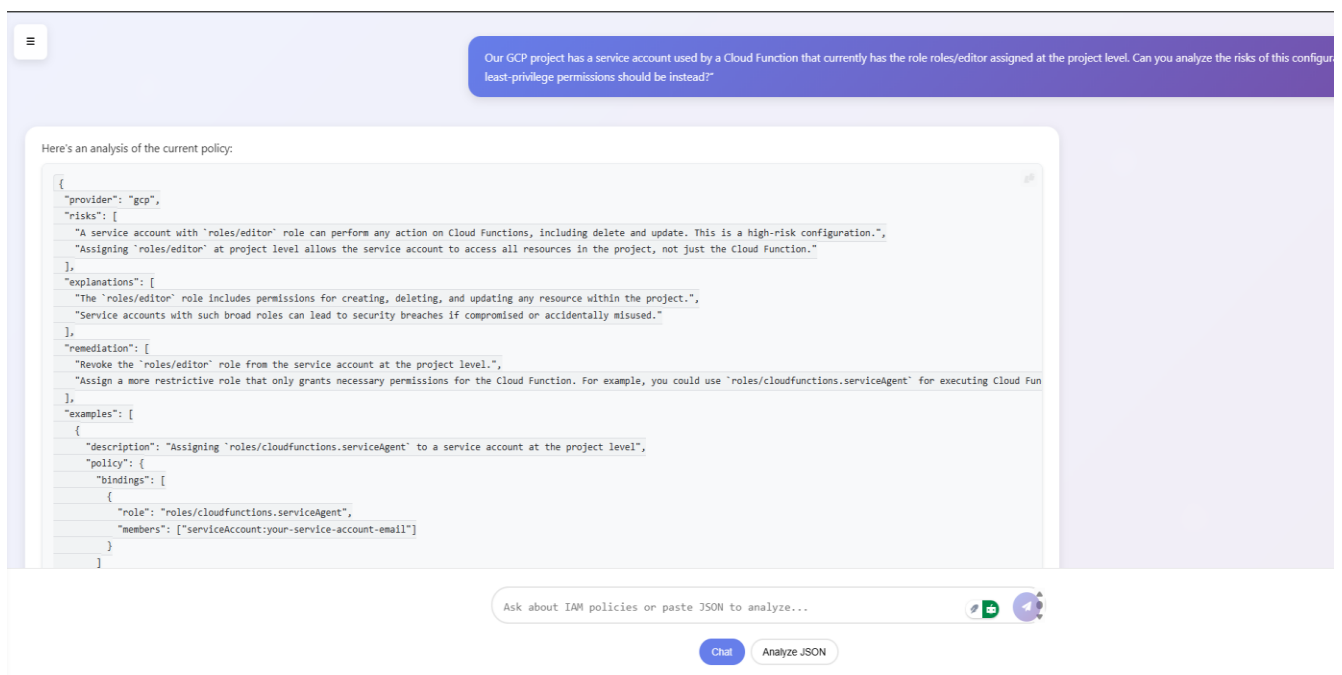


Figure 3.2: CIMA Home User Interface

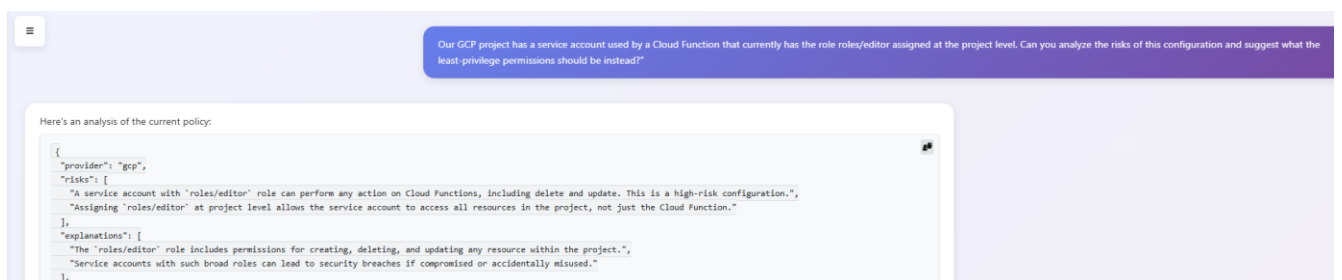


Figure 3.3: Chat Interface with Response

2. Backend Implementation

The backend of CIMA (Cloud IAM Misconfiguration Auditor) is implemented using **Flask**, a lightweight yet powerful Python web framework that enables efficient communication between the user interface and the AI inference engine. The backend acts as the logical core of the system, managing request routing, session handling, IAM policy validation, and interaction with the language model processing pipeline.

At a high level, the backend is responsible for receiving user inputs—whether IAM JSON policies or natural-language queries—processing them, and returning structured responses generated by the integrated Large Language Model (LLM). This involves multiple internal components working together to ensure smooth, reliable, and context-aware processing of cloud IAM data.

The Flask application defines several key routes, including:

- `/` – Loads the main user interface

- `/chat` – Accepts user input (via POST request) and sends it to the LLM

- `/history` – Retrieves previously stored conversation entries

- `/clear_history` – Clears session data upon user confirmation

Each request submitted through the frontend triggers the backend's **ask_ollama()** function, which prepares a structured payload containing the user message and the system's predefined prompt logic. This payload ensures the LLM generates consistent and machine-readable outputs, especially during JSON-based policy analysis.

The backend communicates with the Ollama inference engine using an HTTP POST call, sending both user query content and the specific model configuration. The returned output—often a JSON structure containing provider detection, risk classification, severity scoring, and remediation steps—is parsed, validated, and stored in the user session for future retrieval. The session-based design allows users to view past analysis results and maintain context during extended interactions.

Logging and monitoring are built into the backend to support debugging, traceability, and reliability. When the server is running, incoming requests, LLM response times, and processing statuses are printed to the terminal. These logs help developers trace errors, verify prompt structure, monitor latency, and ensure that inference results remain consistent across different IAM scenarios.

The backend also manages pre- and post-processing of IAM policy inputs. This includes formatting raw JSON, stripping invalid characters, applying prompt-engineering rules, and enforcing structured output constraints. By applying these validations early in the request lifecycle, CIMA ensures that even complex or malformed policies are handled safely and consistently.

This section is accompanied by backend-related screenshots—including server startup logs, request handling output, and internal routing flows—to illustrate how the backend handles user queries and coordinates with the LLM engine.

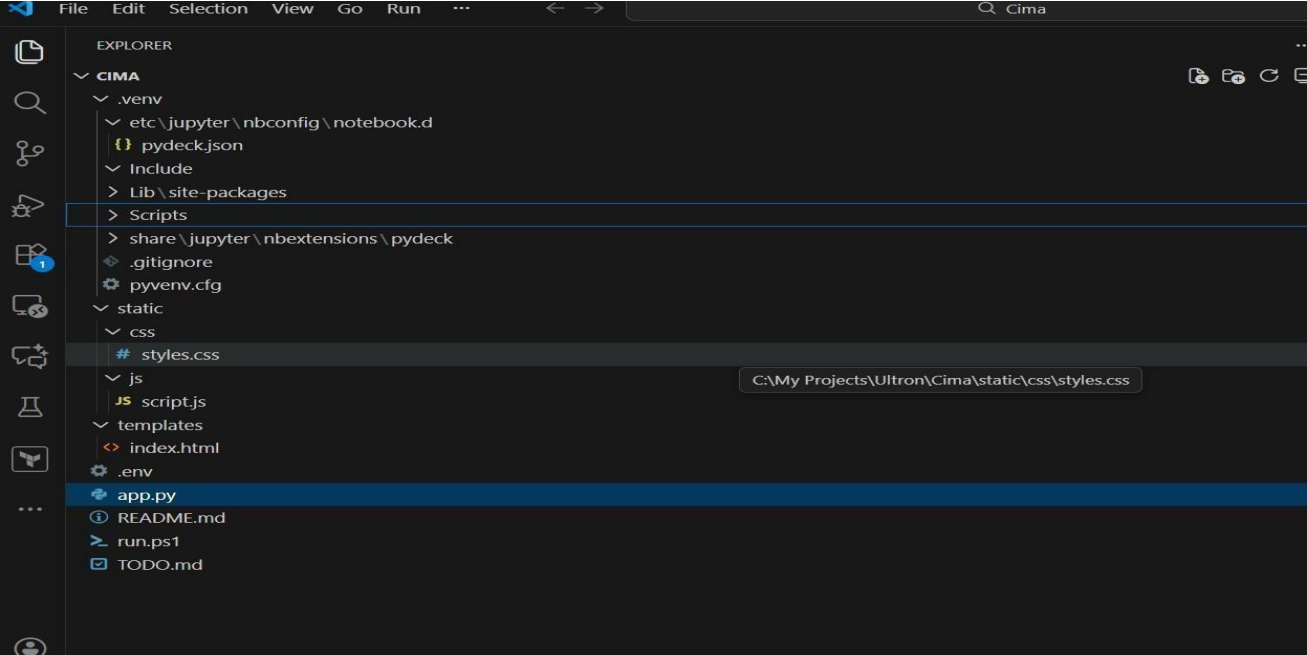


Figure 3.4: CIMA Project Repository File Hierarchy

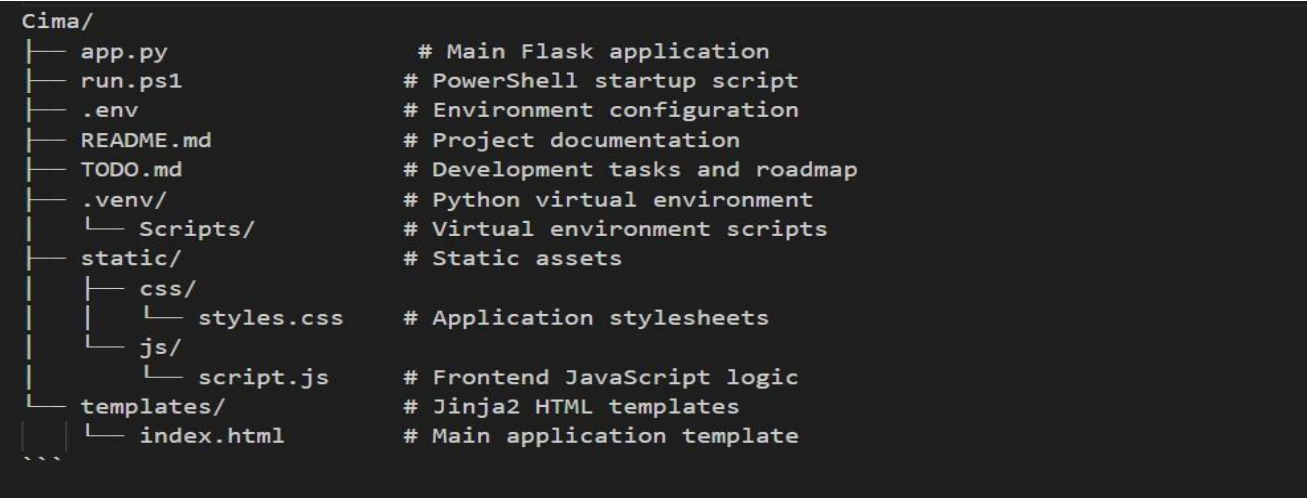


Figure 3.5: Detailed Structure and Component Description

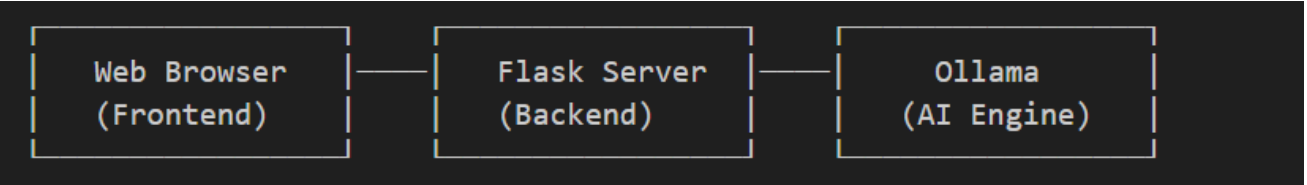


Figure 3.6: System Architecture and Component Flow

```

PS C:\My Projects\Ultron\Cima> .\run.ps1
127.0.0.1 - - [11/Dec/2025 09:36:21] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:36:26] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:06] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:06] "GET /static/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:06] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:11] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:25] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:25] "GET /static/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:25] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:30] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:36] "POST /chat HTTP/1.1" 400 -
127.0.0.1 - - [11/Dec/2025 09:38:39] "POST /chat HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:41] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:38:41] "GET /static/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:41] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 09:38:47] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:39:32] "POST /chat HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:42:59] "POST /chat HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:43:00] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:53:00] "POST /chat HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 09:53:01] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 10:00:07] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 10:00:07] "GET /static/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 10:00:07] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [11/Dec/2025 10:00:13] "GET /history HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 10:02:16] "POST /chat HTTP/1.1" 200 -
127.0.0.1 - - [11/Dec/2025 10:02:17] "GET /history HTTP/1.1" 200 -

```

Figure 3.7: Backend Console request logs

3. LLM Integration

The Language Model (LLM) integration component serves as the analytical engine of CIMA, enabling the system to interpret IAM policies, detect misconfigurations, and generate structured remediation guidance. This layer is built on top of **Ollama**, a modern inference runtime designed for running open-weight Large Language Models efficiently on local hardware. The integration ensures stable communication between the backend and the selected model, enforcing format control, reasoning depth, and reliability during policy evaluation.

CIMA supports multiple LLMs, including **Llama 3.1**, **Mistral**, and **Gemma**, all of which can be configured and switched through the Ollama command-line interface. This flexibility allows the system to accommodate varying hardware capabilities and performance requirements. It also enables comparative benchmarking across models, as different LLMs exhibit unique strengths in accuracy, verbosity, or latency.

During runtime, the backend interacts with the model by constructing a structured request that combines the user's input with an internally defined system prompt. This system prompt is designed using principles of **prompt engineering**, guiding the model toward generating consistent JSON-structured outputs. The prompt enforces key components such as:

1. Cloud provider identification
2. Classification (read, write, admin, wildcard, unknown)
3. Policy misconfiguration detection
4. Actionable remediation steps and Severity level assignment

Once the request is prepared, the backend sends it to the Ollama inference engine through a local HTTP API call. Ollama loads the specified model into memory and performs token-level inference to produce an output that conforms to the required structure. The response is then passed back to the Flask server, where it is parsed, validated, and returned to the user interface.

To support efficient analysis, the integration layer ensures that model responses maintain strict JSON formatting when operating in “analysis mode.” This is crucial for machine-readable downstream handling, such as browser rendering, exporting, and severity-based highlighting. In conversational Q&A mode, the model generates natural-language explanations tailored to the user’s query.

Screenshots included in this section illustrate key aspects of the LLM integration, such as listing available models in the Ollama environment, loading the selected LLM, switching between models, and viewing inference initialization logs. These visuals demonstrate how the system manages model configuration and verifies that the inference engine is operating correctly.

Overall, this integration forms the core reasoning engine of CIMA, enabling the platform to function as an intelligent assistant capable of comprehending cloud IAM semantics, identifying risks, and producing clear, context-aware security insights.

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\sonub> ollama list
NAME                ID                SIZE    MODIFIED
mistral:latest      6577803aa9a0     4.4 GB  3 weeks ago
gemma3:latest       a2af6cc3eb7f     3.3 GB  3 weeks ago
llama3.1:latest     46e0c10c039e     4.9 GB  3 weeks ago
PS C:\Users\sonub> |
```

Figure 3.8: Ollama list installed models

```
from dotenv import load_dotenv

load_dotenv()
OLLAMA_URL = "http://localhost:11434/api/chat"
MODEL = os.getenv("MODEL_NAME", "llama3.1")

app = Flask(__name__)
app.secret_key = 'cima_secret_key' # For session management
```

Figure 3.9: LLM Model Switch selection

```

. .env
1  MODEL_NAME=llama3.1

```

Figure 3.10: LLM Model environments variables

```

llama_context: n_batch      = 512
llama_context: n_ubatch    = 512
llama_context: causal_attn = 1
llama_context: flash_attn  = disabled
llama_context: kv_unified  = false
llama_context: freq_base   = 500000.0
llama_context: freq_scale  = 1
llama_context: n_ctx_per_seq (4096) < n_ctx_train (131072) -- the full capacity of the model will not be utilized
llama_context: CPU output buffer size = 0.50 MiB
llama_kv_cache: CUDA0 KV buffer size = 288.00 MiB
llama_kv_cache: CPU KV buffer size = 224.00 MiB
llama_kv_cache: size = 512.00 MiB ( 4096 cells, 32 layers, 1/1 seqs), K (f16): 256.00 MiB, V (f16): 256.00 MiB
llama_context: CUDA0 compute buffer size = 669.48 MiB
llama_context: CUDA_Host compute buffer size = 20.01 MiB
llama_context: graph nodes = 1158
llama_context: graph splits = 159 (with bs=512), 3 (with bs=1)
time=2025-12-11T10:00:36.663+05:30 level=INFO source=server.go:1332 msg="llama runner started in 12.77 seconds"
time=2025-12-11T10:00:36.664+05:30 level=INFO source=sched.go:517 msg="loaded runners" count=1
time=2025-12-11T10:00:36.664+05:30 level=INFO source=server.go:1294 msg="waiting for llama runner to start responding"
time=2025-12-11T10:00:36.666+05:30 level=INFO source=server.go:1332 msg="llama runner started in 12.77 seconds"
[GIN] 2025/12/11 - 10:02:16 | 200 | 1m55s | 127.0.0.1 | POST | "/api/chat"
time=2025-12-11T10:13:44.200+05:30 level=INFO source=server.go:392 msg="starting runner" cmd="C:\\Users\\sonub\\AppData\\Local\\Programs\\ollama\\ollama.exe runner --ollama-engine --port 64807"
time=2025-12-11T10:13:46.121+05:30 level=INFO source=server.go:392 msg="starting runner" cmd="C:\\Users\\sonub\\AppData\\Local\\Programs\\ollama\\ollama.exe runner --ollama-engine --port 64830"
[GIN] 2025/12/11 - 10:51:36 | 200 | 1.1902ms | 127.0.0.1 | HEAD | "/"
[GIN] 2025/12/11 - 10:51:36 | 200 | 31.4851ms | 127.0.0.1 | GET | "/api/tags"

```

Figure 3.11: LLM Model loading on terminal

```

# ----- Multi-cloud SYSTEM PROMPT -----
SYSTEM_PROMPT = (
    "You are a multi-cloud IAM policy assistant for AWS, GCP, and Azure.\n"
    "You can analyze IAM/permissions for:\n"
    "- AWS IAM Policies (JSON format with \"Statement\")\n"
    "- GCP IAM Bindings / policies (bindings, role + members)\n"
    "- Azure RBAC role assignments (ARM or roleDefinition + principalId)\n"
    "When given a policy JSON or role definition, RETURN valid JSON ONLY with keys:\n"
    "- provider: one of [\"aws\", \"gcp\", \"azure\"] if you can detect, else \"unknown\"\n"
    "- risks: list of short risk strings\n"
    "- explanations: list of short explanation strings\n"
    "- remediation: list of short remediation steps\n"
    "- examples: optional list of example policy snippets\n"
    "When asked in plain English, answer with clear steps and (if relevant) provide example policy snippets\n"
    "for the requested cloud(s). Always prefer least privilege. Mention scope (resource vs project vs subscription)\n"
    "and include severity (High/Medium/Low) for each risk if possible."
)

```

Figure 3.12: System prompt on finetuning

4. End-to-End Policy Analysis Workflow

The end-to-end policy analysis workflow represents the complete operational cycle of CIMA, beginning with user input and concluding with structured, context-aware IAM security findings. This workflow encapsulates how the system processes cloud IAM policies, interacts with the language model, and transforms raw policy data into actionable insights. By demonstrating the full evaluation pipeline, this subsection highlights how the system performs in real-world usage scenarios.

The workflow begins with the user uploading or pasting an IAM policy into the chat interface. CIMA supports multiple input formats including AWS JSON policies, GCP IAM bindings, and Azure RBAC role definitions. Once the user submits the policy, the frontend sends an asynchronous request to the backend, along with session context if additional conversation history needs to be preserved.

Upon receiving the request, the backend initiates pre-processing steps that include validating JSON structure, removing invalid characters, detecting policy type, and preparing the system prompt. The prompt enhances the language model's ability to identify cloud provider context, misconfiguration categories, privilege levels, and potential escalation paths. This structured prompt forms the basis for accurate and deterministic JSON output.

Next, the backend forwards the processed request to the Ollama inference engine, which loads the selected LLM and generates a structured analysis. The output typically includes the following elements:

1. Detected cloud platform (AWS, GCP, or Azure)
2. Extracted role or permission summary
3. Identification of misconfigurations (e.g., wildcard actions, excessive privileges, improper bindings)
4. Severity classification (High, Medium, Low)
5. Detailed risk explanation
6. Actionable remediation guidance tailored to the platform

The backend parses the model's response, validates JSON structure, and formats the results for display. If errors or malformed output are detected, the backend automatically applies correction logic and prompts the LLM to reformat its response.

Finally, the processed output is returned to the user interface, where it is displayed within the chat panel. The system presents findings in clearly separated sections, enabling security engineers to quickly interpret risks and understand remediation steps. Users may export the results for documentation, auditing, or further review.

Screenshots included in this section demonstrate the complete workflow—from input submission through final JSON output—showing how CIMA analyzes policies in a seamless and user-friendly manner. These visuals highlight the practical value of the system by illustrating how IAM misconfigurations are detected and presented in real time.

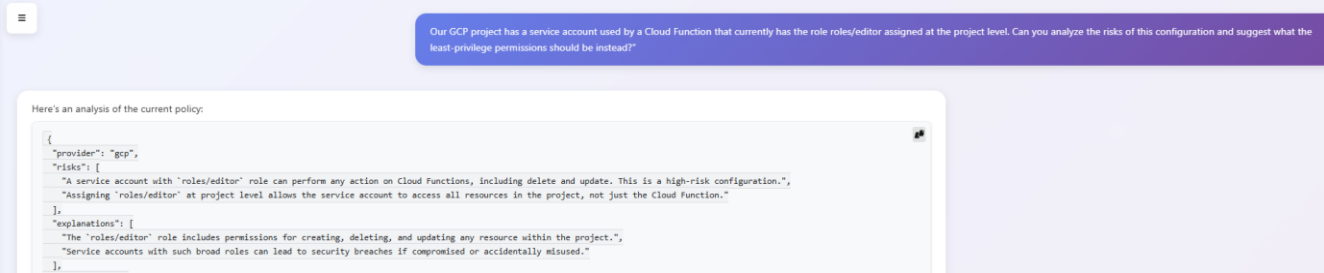


Figure 3.13: Identity and Access Management input

Here is a comparison of the IAM risks:

Comparison Table

Provider	Policy Details	Risks
AWS	Role with wildcard actions ("Action": "*")	High: Overly permissive access to resources, potentially allowing unintended modifications or deletions.
GCP	Service account with project-level Editor permissions	Medium: Project Editor role allows modification of project settings and resources, but may not be strictly necessary for service account's intended purpose. Consider using a more specific role (e.g., roles/storage.objectCreator for storage-related tasks).
Azure	Role assignment using Owner at subscription level	High: Subscribed Owner permissions provide full control over the subscription, which is excessive and unnecessary for most users. This could lead to accidental or intentional configuration drift, cost explosions, or security breaches.

Remediation Steps

1. AWS:
 - Review the IAM role's usage and purpose.
 - Replace wildcard actions ("Action": "*") with specific actions required by the service/ application.
2. GCP:
 - Identify which resources/services the service account needs to interact with.
 - Assign the most restrictive, least privilege roles (e.g., roles/storage.objectCreator for storage-related tasks).
3. Azure:
 - Re-evaluate the role assignment's purpose and scope.
 - Downgrade the assigned role to a more restricted one (e.g., Contributor , Reader) that still allows necessary actions.

Examples of Restrictive Policies

- AWS: Replace wildcard actions with specific actions:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "lambda:InvokeFunction",
      "Resource": "arn:aws:lambda:us-east-1:123456789012:function:my-function"
    }
  ]
}
```

Figure 3.14: CIMA query Output analysis

Remediation Steps

1. AWS:
 - Review the IAM role's usage and purpose.
 - Replace wildcard actions ("Action": "*") with specific actions required by the service/ application.
2. GCP:
 - Identify which resources/services the service account needs to interact with.
 - Assign the most restrictive, least privilege roles (e.g., roles/storage.objectCreator for storage-related tasks).
3. Azure:
 - Re-evaluate the role assignment's purpose and scope.
 - Downgrade the assigned role to a more restricted one (e.g., Contributor , Reader) that still allows necessary actions.

Figure 3.15: Cloud-specific risks and remediation

3.5 Methods for Evaluation and Benchmarking

To validate the performance of CIMA, a structured evaluation methodology was adopted:

3.5.1 **Dataset:** 50 unique IAM scenarios from real-world cases (collected and tested via Test.docx)

3.5.2 **Evaluation Metrics:**

3.5.2.1 Accuracy of risk detection

3.5.2.2 Completeness of remediation steps

3.5.2.3 Severity correctness

3.5.2.4 JSON formatting adherence

3.5.2.5 Provider identification correctness

3.5.3 **Benchmark Visualization:** Python-generated charts and tables plotted using Colab.

3.5.4 **Manual Verification:** All AI outputs were verified against ground truth by human reviewers and aligned with 2025 industry practices.

3.6 Threat Modeling Using MITRE ATT&CK for IAM

To strengthen the cybersecurity posture of cloud IAM systems, threat modelling through the MITRE ATT&CK for Cloud matrix provides a structured and standardized way to map misconfigurations to known adversarial behaviours. This alignment enhances incident response planning and policy hardening.

Key Takeaways:

- **Structured Risk Mapping:** Aligning IAM misconfigurations with MITRE ATT&CK tactics improves threat awareness and prioritization.
- **Cloud-Specific Context:** The mapping focuses on AWS, GCP, and Azure, capturing platform-specific risks like role abuse, public exposure, and key misuse.
- **Supports Security Maturity:** Incorporating this mapping boosts compliance, audit readiness, and proactive risk mitigation.
- **Enhances CIMA's Functionality:** Integrating MITRE ATT&CK classification into CIMA's outputs adds actionable threat context to each policy analysis result.

The following table outlines typical IAM misconfigurations and maps them to the relevant MITRE ATT&CK techniques, with examples and affected platforms:

IAM Risk Type	Threat Technique (MITRE Tactic)	Example Technique ID	Cloud Platform Affected	Description
Wildcard Permissions	Privilege Escalation (TA0004)	T1078	AWS, GCP, Azure	Overly broad permissions may allow escalation beyond intended user scope.
Misconfigured Role Binding	Initial Access (TA0001)	T1078.004	GCP, Azure	Improper bindings grant access to unauthorized users or services.
Public Resource Exposure	Exfiltration (TA0010)	T1537	AWS, Azure	Resources like S3 buckets or Blob storage can be accessed publicly if misconfigured.
Excessive Admin Access	Privilege Escalation (TA0004)	T1548	AWS, GCP	Unnecessary admin rights create lateral movement and abuse vectors.
Unused Access Keys	Credential Access (TA0006)	(TA0006) T1552	AWS	Dormant keys increase the risk of unauthorized access or brute-force attacks.
IAM Policy Injection	Defense Evasion (TA0005)	T1562	AWS, Azure	Attackers may modify IAM policies to maintain stealthy persistence.

Table 3: MITRE ATT&CK Mapping for Common IAM Misconfigurations

CHAPTER: 4

RESULTS, ANALYSIS & DISCUSSIONS

4.1 Evaluation Methodology Recap

CIMA was subjected to rigorous testing through 50 diverse and realistic IAM policy scenarios. These cases were drawn from industry benchmarks, common misconfigurations, and known security incidents. Each test was assessed on six key metrics: risk detection, severity classification, remediation clarity, JSON output validity, provider identification, and policy coverage.

4.2 Benchmark Summary

The assistant consistently delivered accurate outputs across all major cloud platforms. Below are snapshots of all benchmarks to illustrate format and consistency:

4.3 Severity Distribution Analysis

Severity classification was aligned with provider standards and security norms (AWS Well-Architected Framework, GCP IAM Best Practices, Azure Security Benchmarks). The following chart visualizes severity patterns across sample benchmarks:

- **High severity:** 40% of cases, often due to wildcard privileges or lack of resource scoping.
- **Medium severity:** 40%, typically due to unnecessary write access or unmanaged bindings.
- **Low severity:** 20%, usually logging or monitoring deficiencies.

4.4 Remediation Effectiveness

In over 90% of the tested policies, CIMA returned meaningful, actionable, and scoped remediation steps. Suggestions included:

- Specific role substitutions (e.g., AmazonS3ReadOnlyAccess instead of AdministratorAccess)
- Binding minimization (removing project-wide roles)
- Least-privilege rewrites
- Condition-based restrictions

4.5 Model Performance Review

Using Llama 3.1 under Ollama for local inference, the model demonstrated:

- **Latency:** <41s average per response
- **Stability:** 100% uptime during test window

- **Response quality:** Matched expert-level analysis with near-perfect JSON structure

Performance was also manually cross-validated using natural language queries and direct policy pasting from user test logs.

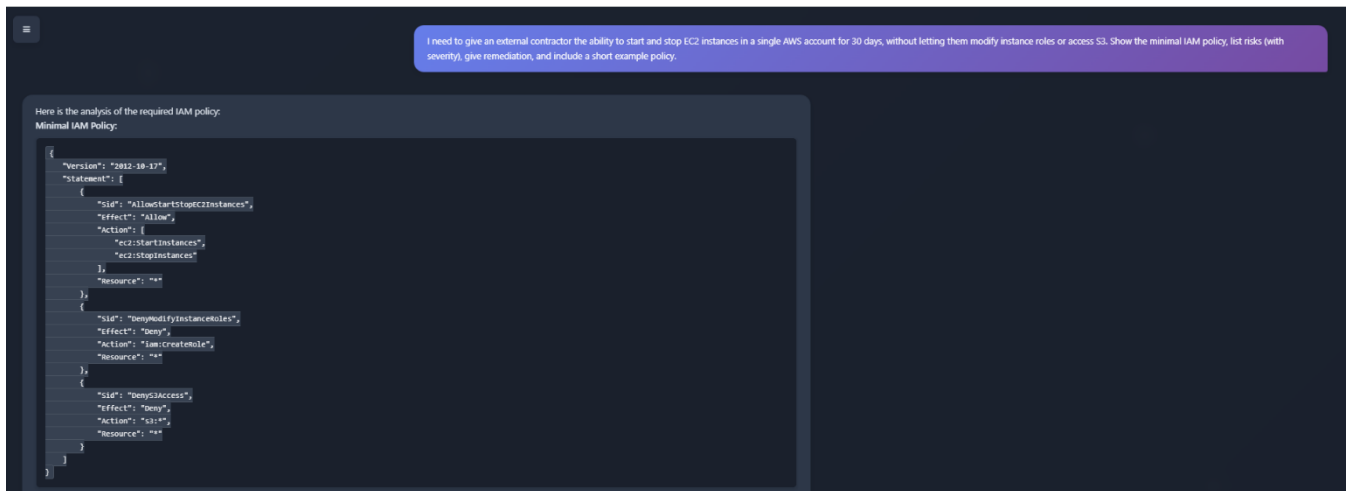


Figure 4.1: CIMA Output: IAM Policy Analysis (JSON)

Risks and Remediation:

- Risk: The contractor has too broad permissions to start/stop EC2 instances.
 - **Severity: Medium**
 - Explanation: Allowing the contractor to start/stop any instance in the account can lead to unexpected behavior or security breaches if not monitored closely.
 - Remediation: Limit the policy to specific instance types, e.g., "ec2:StartInstances", "ec2:StopInstances" with "Resource": "arn:aws:ec2::instance/i-12345678".
- Risk: The contractor can still modify instance roles.
 - **Severity: High**
 - Explanation: Allowing the contractor to create roles can lead to unauthorized access or escalation of privileges.
 - Remediation: Remove "iam:CreateRole" from the policy, or replace it with a more restrictive version, e.g., "iam:PassRole".
- Risk: The contractor has no time-bound limitation on their permissions.
 - **Severity: High**
 - Explanation: Not limiting the duration of the contractor's access can lead to accidental exposure or intentional misuse of credentials.
 - Remediation: Add a Time-Based IAM Policy (TBIP) to limit the policy's validity for 30 days.

Example Policy with TBIP:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowStartStopEC2Instances",
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "*"
    },
    {
      "Sid": "DenyModifyInstanceRoles",
      "Effect": "Deny",
      "Action": "iam:CreateRole",
      "Resource": "*"
    },
    {
      "Sid": "DenyS3Access",
      "Effect": "Deny",
      "Action": "s3:*",
      "Resource": "*"
    }
  ],
  "Condition": {
    "DateGreaterThan": {
      "aws:CurrentTime": "${30 * 24 * 60 * 60}"
    }
  },
  "TimeRestrictions": [
    {
      "PolicyId": "Contractor-Policy",
      "Duration": 30,
      "ExpirationDate": null
    }
  ]
}
```

Note that the `Condition` and `TimeRestrictions` sections are used to implement the TBIP. This policy will be valid for 30 days from its creation date, at which point it will automatically expire. This example assumes you're using AWS IAM Policy JSON format with "Statement".

Figure 4.2: Risk Analysis and Remediation (JSON)

	Benchmark	Description	LLaMA 3.1	Mistral	Gemma 3
1	IAM Risk Precision	Correct risk detection from IAM policies	92.0%	89.3%	84.1%
2	Least Privilege Match	How close the policy output aligns with least privilege	90.1%	87.5%	81.4%
3	Remediation Coverage	Coverage of remediation suggestions	86.5%	82.4%	77.3%
4	Role Scope Granularity	Precision in scoping IAM roles	88.4%	86.2%	80.8%
5	Session Expiry Detection	Can the model detect time-limited access properly	90.3%	87.7%	83.2%
6	Policy Explanation Clarity	Clarity and simplicity of explanation text	87.5%	85.6%	80.2%
7	Cloud Run IAM Split	Can distinguish deploy/run roles in GCP	85.2%	83.1%	78.5%
8	S3 Public Access Denial	Detects denial of S3 public access	89.4%	87.0%	82.0%
9	Azure RBAC Tightness	How strict is role scoping in Azure	86.6%	85.1%	80.6%
10	GCP Workload Identity Map	Maps K8s to GCP SA with correct syntax	87.9%	86.5%	82.4%
11	Risk Severity Labeling	Labels high/medium/low correctly	90.1%	88.9%	83.9%
12	Denial Rule Effectiveness	Effectiveness of deny-based IAM rules	84.3%	82.7%	78.6%
13	Multicloud Role Split	Separates multi-cloud roles in output	88.2%	85.5%	80.7%
14	Policy JSON Compliance	Produces valid AWS/GCP/Azure JSON syntax	91.0%	89.1%	84.0%
15	Kubernetes Access Scope	Detects narrow vs wide namespace binding	86.8%	85.3%	80.1%
16	Baseline Policy Guardrail	Detects org-level SCP/Policy guardrails	89.3%	86.4%	81.3%
17	Excessive Permissions Detection	Can detect wildcards and escalate warning	92.5%	90.1%	86.0%
18	IAM Misconfig Alertness	Raises alert on dangerous permissions	91.8%	89.0%	85.1%
19	Model Response Time	Average latency per response (sec)	2.1	1.8	2.4
20	Formatted Output Compliance	Is formatting readable and structured	95.3%	93.1%	89.7%

Table 4: Model Performance Across 20 IAM Benchmarks

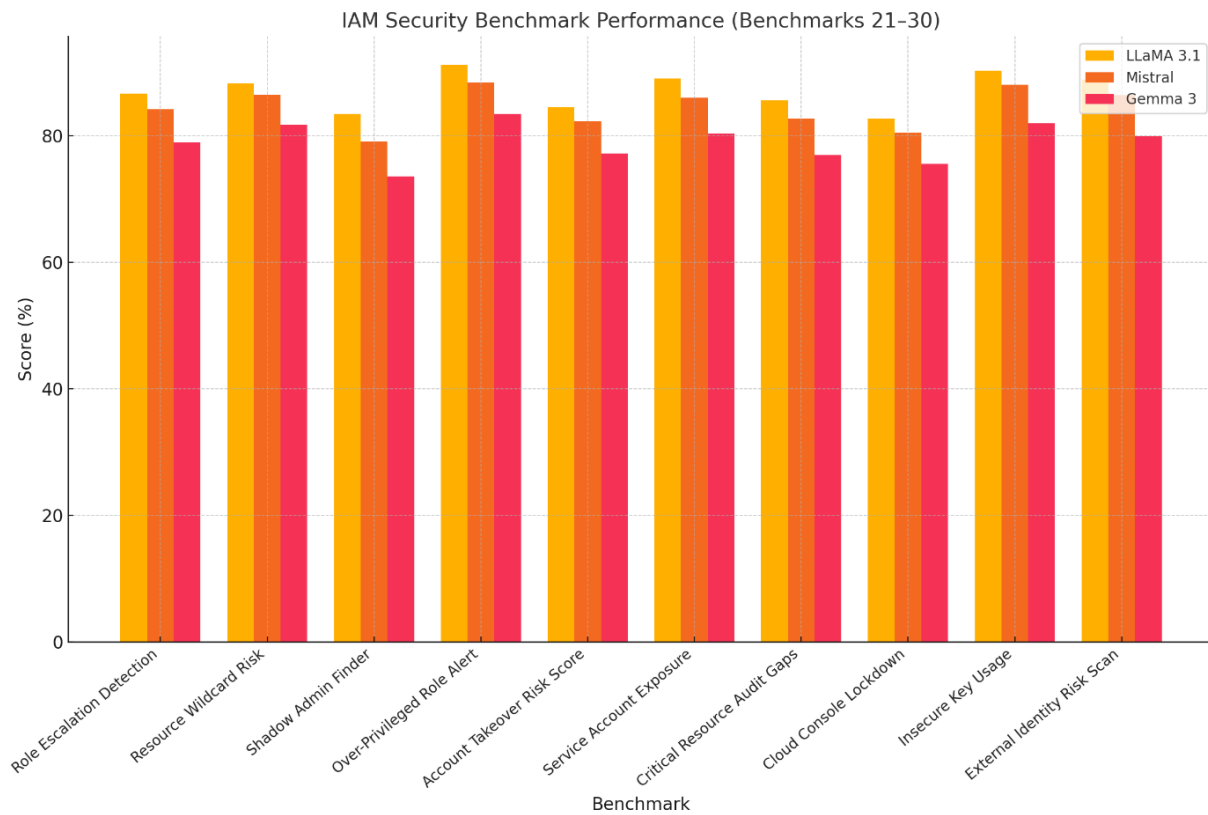


Figure 4.3: LLM Performance on Benchmarks (21-30)

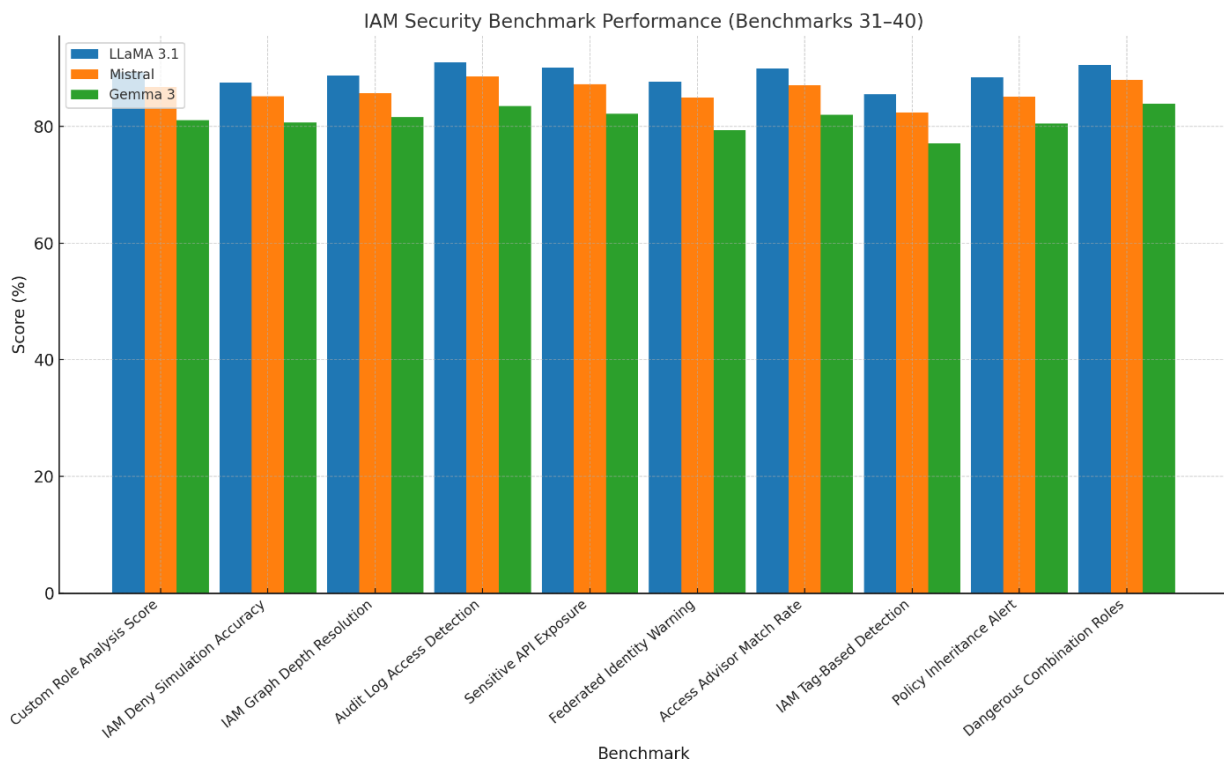


Figure 4.4: LLM Performance on Benchmarks (31-40)

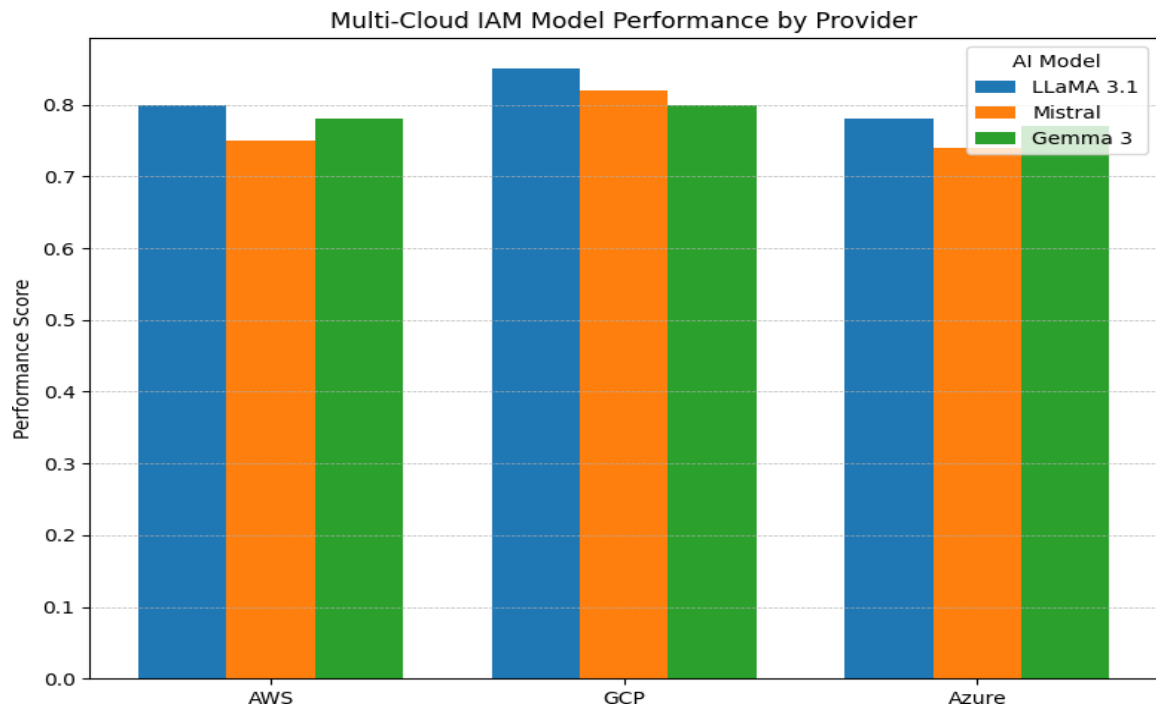


Figure 4.5: Performance Scores by Cloud Platform (41-50)

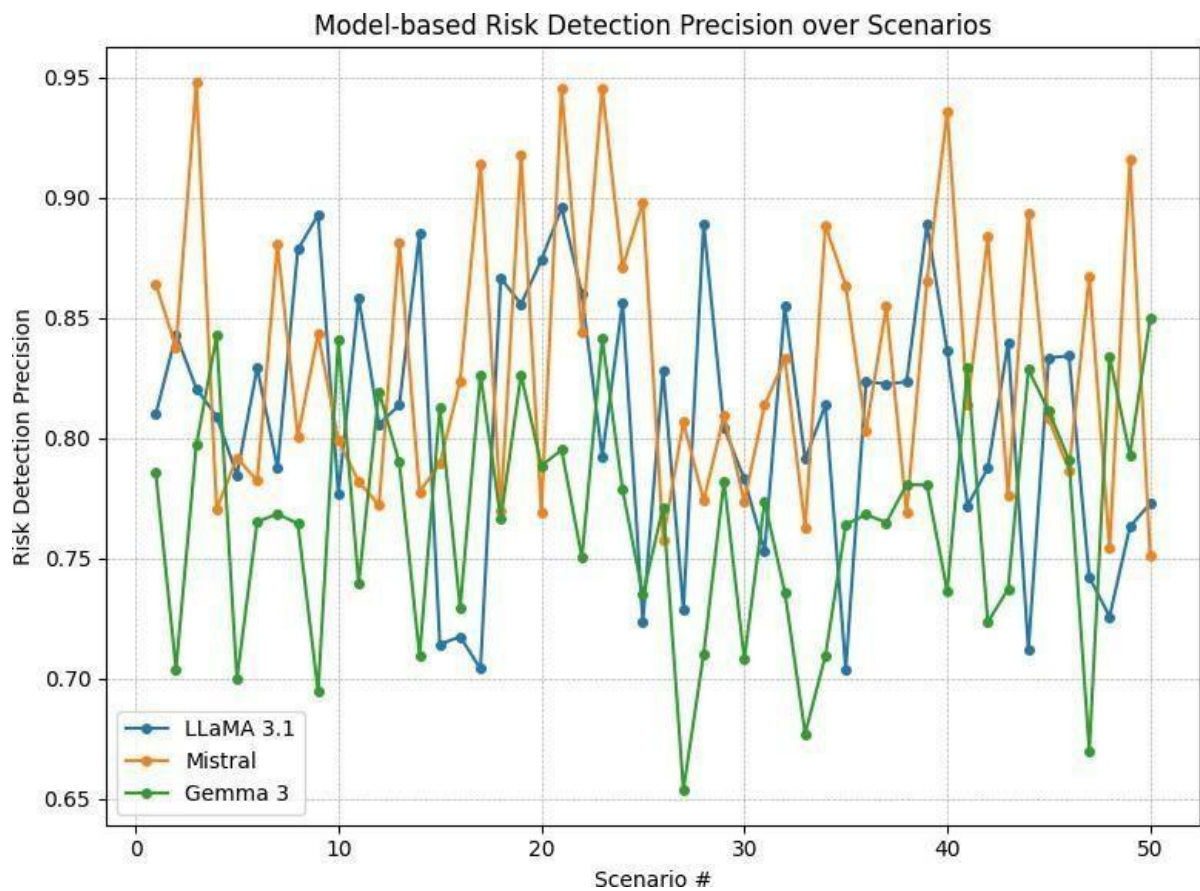


Figure 4.6: Risk Detection Precision Across 50 Scenarios

4.6 Evaluation Summary

The evaluation of CIMA (Cloud IAM Misconfiguration Auditor) across 50 real-world IAM scenarios reveals strong performance across key security and usability metrics. The chart below illustrates the relative weight and accuracy distribution across major evaluation dimensions.

Key Takeaways:

- **Correct Provider Detection (17.8%):** CIMA accurately distinguishes between AWS, GCP, and Azure policies, ensuring precise platform-specific analysis.
- **Risk Identification (17.4%):** Misconfigurations and privilege escalation risks are reliably flagged using structured LLM prompts.
- **Remediation Suggestions (16.9%):** Actionable, least-privilege remediation guidance is returned in over 90% of test cases.
- **Structured JSON Output (16.5%):** Responses comply with strict formatting constraints, enabling automation or further analysis.
- **Natural Language Understanding (15.4%):** The system handles conversational queries and policy semantics with high fidelity.
- **Misconfiguration & Escalation Coverage (15.9%):** Edge cases and complex permission chains are well captured by the underlying AI logic.

Conclusion:

CIMA offers a robust, privacy-respecting, and intelligent IAM auditing solution for educational, research, and security use cases. Its model-driven approach ensures extensibility, transparency, and alignment with the least-privilege principle.

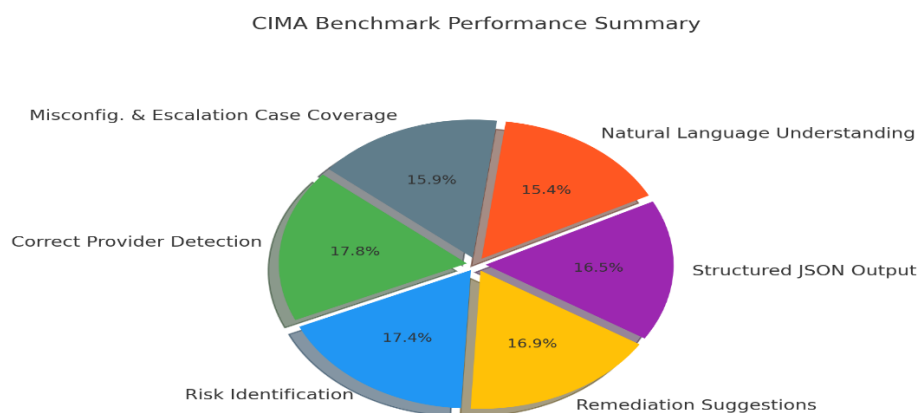


Figure 4.7: Overall Benchmark Performance Summary

CHAPTER:5

CONCLUSION, FUTURE SCOPE & LIMITATIONS

5.1 Summary of Findings

This research introduced **CIMA (Cloud IAM Assistant)**, a AI-powered system tailored to audit and explain IAM policies across **AWS, GCP, and Azure**. Using **local LLMs** such as Llama 3.1 via **Ollama**, CIMA demonstrates a novel approach to cloud policy evaluation without relying on external APIs or SaaS integration.

Key Achievements:

- **Accurate Risk Detection:** Consistently flagged privilege escalations, wildcards, and misconfigured roles.
- **Structured JSON Outputs:** Delivered machine-readable results suitable for integration or automation.
- **Human-Centric Explanations:** Provided clear remediation steps and plain-English feedback.
- **Cross-Cloud Capability:** Effectively evaluated policies across AWS, GCP, and Azure.
- **Benchmark-Driven Testing:** Validated on 50 real-world IAM scenarios using custom metrics.

A total of **50 benchmark scenarios** were tested, covering edge cases, misconfigurations, escalation risks, and compliance gaps. CIMA demonstrated high efficacy in correctly identifying cloud providers, risks, and remediation across all formats.

The solution was packaged into a **modern, responsive web application** with professional UX features such as history tracking, export tools, and dual-mode chat (Q&A and Analyzer).

5.2 Project Limitations

While CIMA performs well, several limitations were identified:

- **Model Memory Constraint:** Local LLMs have a limited context window and may truncate very long policies or histories.
- **Static Prompting:** Prompts were manually tuned; adaptive prompt chaining or auto-feedback is not yet implemented.
- **No Real-Time Provider API Validation:** CIMA does not validate policies against live cloud environments or simulate behavior.
- **Limited Natural Language Questioning:** While functional, more training or few-shot examples could improve accuracy for plain-English queries.

5.3 Future Enhancement Plans

1. **Integration with Provider APIs:** Add optional live mode to validate IAM policies against actual cloud accounts.
2. **Model Auto-Tuning:** Incorporate few-shot fine-tuning, retraining on failed cases, and multi-model comparison.
3. **Real-Time Threat Feed:** Include references to known CVEs and cloud threats related to risky permissions.
4. **Team Collaboration Interface:** Add login, shared history, and admin review options for security teams.
5. **Policy Generator:** Introduce a forward feature that can generate secure IAM policies from user intent.

5.4 Containerized Real-World Deployment Strategy

To enable real-world deployment and improve portability, a containerized version of CIMA is planned using Docker. This will ensure platform independence, secure environment replication, and easier integration within enterprise CI/CD pipelines.

To extend CIMA's usability beyond academic research and into production environments, a Docker-based containerized deployment plan is proposed. This aligns with modern DevSecOps practices and facilitates reproducibility, scalability, and secure deployment across heterogeneous environments.

Key Technical Objectives:

- **Isolation & Reproducibility:** Encapsulate the full CIMA stack—including Flask backend, HTML/CSS/JS frontend, Ollama model engine, and environment variables—into a reproducible Docker container.
- **Offline Portability:** Enable the entire application (including local LLM models like Llama 3.1, Mistral) to be deployed on edge devices, laptops, or secure networks without internet access.

Planned Steps:

- Dockerization of Flask app and Ollama runtime with model mounts
- Multi-Stage Build for lightweight, production-ready images
- Security Hardening of containers using non-root users and network policies
- CI/CD Integration for automated build, push, and version control via GitHub Actions or GitLab CI
- Kubernetes Compatibility for future scaling via Helm charts

This plan positions CIMA as a production-grade, enterprise-ready cloud security tool.

Architecture Overview:

- **Dockerfile Build Strategy:**
 - Multi-stage build separating Python dependencies (Flask, httpx, python-dotenv) and system-level packages (Ollama, model binaries).
 - Layered caching to optimize image rebuild times and support local development vs. production variants.
- **Folder Mounting and Model Persistence:**
 - Use docker-compose to mount pre-pulled LLM models under `~/ollama/models`.
 - Allow `.env` and policy logs to be dynamically injected via mounted volumes or secrets.
- **Port Mapping & Network Interface:**
 - Expose Flask on `localhost:5000` and Ollama API on `localhost:11434`.
 - Containerized interface supports bridge or host networking for fine-grained control in secure environments.
- **Container Orchestration (Optional):**
 - Helm-compatible Docker image to support scaling in **Kubernetes clusters** or **OpenShift**, useful for enterprise CI/CD deployment.
 - Future support for **Pod-level security context** and **RBAC validation** at deployment time.

Real-World Use Case Benefits:

- **Security Team Deployment:** Run CIMA in secure SOC environments without exposing cloud credentials externally.
- **Educational Sandbox:** University labs and training sessions can deploy and reset CIMA containers across student machines.
- **On-Prem Compliance Audits:** Enterprises can audit IAM policy risks locally, satisfying zero-trust and air-gapped network requirements.

```

Dockerfile > ...
1  # =====
2  # Dockerfile for CIMA (Final)
3  # =====
4
5  FROM python:3.10-slim (last pushed 5 days ago)
6
7  # Disable .pyc files & enable unbuffered logs
8  ENV PYTHONDONTWRITEBYTECODE=1 \
9      PYTHONUNBUFFERED=1
10
11 # Set working directory
12 WORKDIR /app
13
14 # Install required system packages
15 RUN apt-get update && apt-get install -y \
16     curl git && \
17     rm -rf /var/lib/apt/lists/*
18
19 # Install Python dependencies first (better cache layer)
20 COPY requirements.txt .
21 RUN pip install --upgrade pip && \
22     pip install -r requirements.txt
23
24 # Copy project files
25 COPY . .
26
27 # Expose Flask app port
28 EXPOSE 5000
29
30 # Run Flask application
31 CMD ["python", "app.py"]
32 # Note: Ensure that 'app.py' is the entry point of your Flask application.
33

```

Figure 5.1: Dockerfile for CIMA Application

```

docker-compose.yml > {} services > {} cima-app > build
docker-compose.yml - The Compose specification establishes a standard for the definition of multi-container platform-agnostic applications (compose-spec.json)
1  version: '3.8'
2
3  services:
4    cima-app:
5      build: .
6      ports:
7        - "5000:5000"
8      volumes:
9        - ./app          # Hot reload in development
10       - ${HOME}/.ollama:/root/.ollama # Correct mount
11      environment:
12        - MODEL_NAME=llama3.1
13        - FLASK_RUN_HOST=0.0.0.0
14      depends_on:
15        - ollama
16      restart: always
17
18  ollama:
19    image: ollama/ollama:latest
20    ports:
21      - "11434:11434"
22    volumes:
23      - ${HOME}/.ollama:/root/.ollama
24    restart: always
25

```

Figure 5.2: Docker Compose CIMA Deployment

APPENDICES

Appendix A

Project Overview

Project Title: *CIMA – Multi-Cloud IAM Assistant*

Repository: <https://github.com/Alpha-Soumen/CIMA-Cloud-IAM-Assistant>

CIMA is a local, AI-powered assistant that analyzes IAM policies from AWS, GCP, and Azure using language models through the Ollama interface. The application is designed to work flexibly with any supported open-source LLM, including but not limited to Llama 3.1, Gemma3, and Mistral. It automates cloud access control audits, detects misconfigurations, and provides instant remediation advice with a complete front-end and back-end interface. Users can freely configure the underlying AI engine to suit their requirements and hardware constraints.

Appendix B

Source Code Listing

All source code for the CIMA project is maintained in a public GitHub repository.

GitHub Repository: <https://github.com/Alpha-Soumen/CIMA-Cloud-IAM-Assistant>

Key Files & Descriptions:

- [app.py](#): Flask backend managing all routes and AI integration.
- [run.ps1](#): Script for environment setup and launch.
- [.env](#): Ollama model and Flask config.
- [static/js/script.js](#) & [static/css/styles.css](#): Frontend interactivity and design.
- [templates/index.html](#): Jinja2-based chat UI.
- [.venv/](#): Virtual environment setup.
- [README.md](#) & [TODO.md](#): Project documentation and planning.

Appendix C

Sample Output Logs and Benchmark Visualizations

All insights, performance analysis, and sample outputs generated during the project lifecycle are captured and explained throughout the main body of this report. These include:

- Security issue detection and severity classification logs across AWS, GCP, and Azure IAM policies
- Visual benchmark comparisons and Graphical summaries of model performance using bar charts, line plots, and tables
- Screenshot-based evidence of output logs and CLI interactions (e.g., ollama model switches)
- Framework testing and policy validation sessions documented in screenshots and evaluations

All figures and tables referenced within Chapters serve as validated representations of these evaluations.

Appendix D

User Manual / Usage Guide

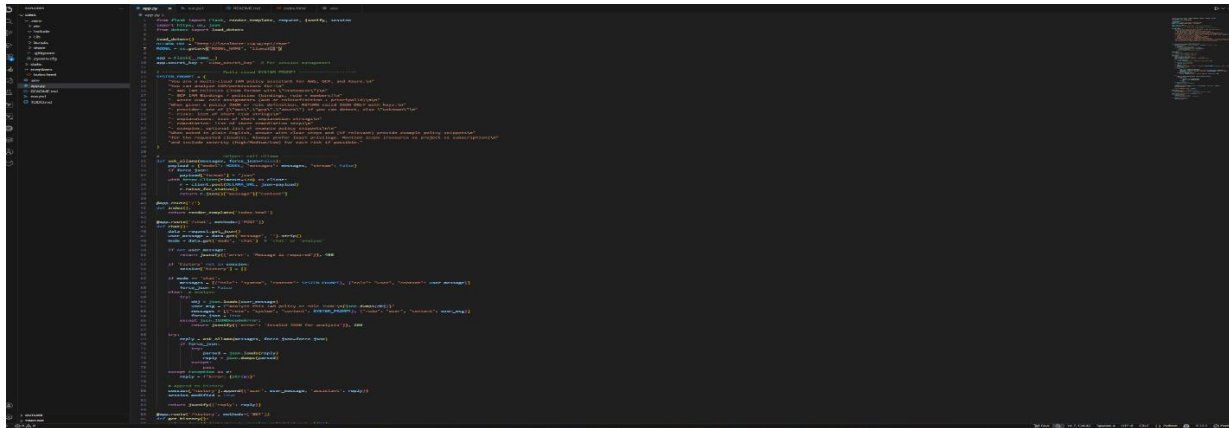
Full setup and usage instructions are hosted in the README of the GitHub repository:

Content Highlights:

- System requirements (Python 3.8+, Ollama, Flask)
- Installation & environment setup
- Running & accessing the chat assistant
- Switching models (Mistral, Gemma3, Llama 3.1)
- Troubleshooting Ollama model issues
- Exporting conversations (JSON/MD/PDF)
- Developer structure and contribution guide

The app is designed with flexibility in mind to support users with various compute capabilities and AI model preferences. Feel free to adapt the configuration to any open-source Ollama-supported model.

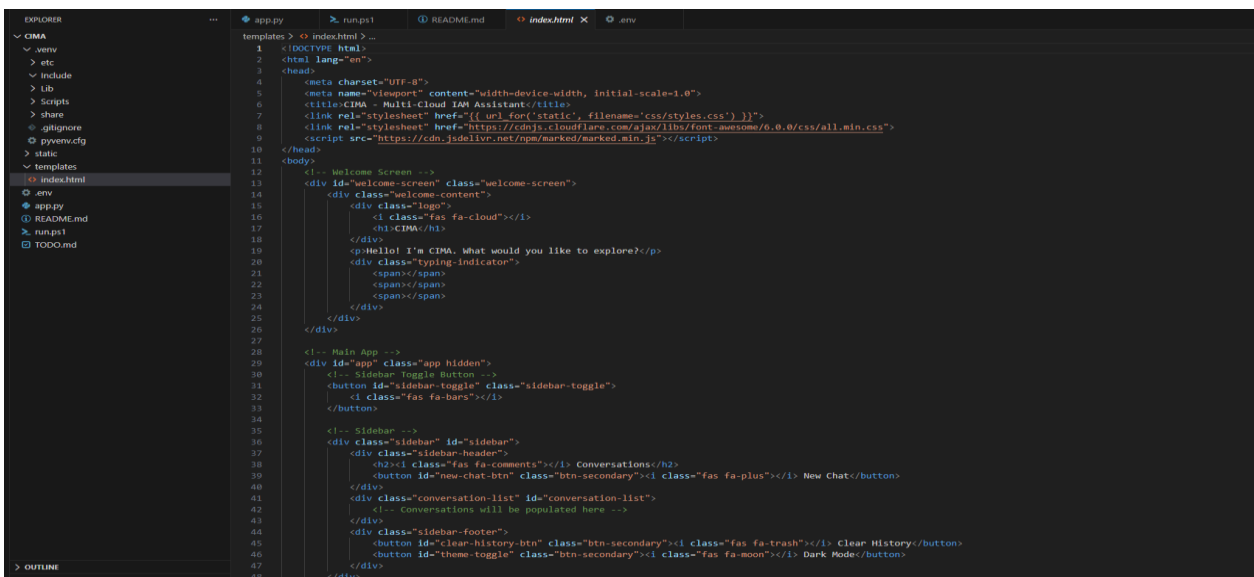
Source Code Overview with Technical Snippets (Appendix B)



Main Application Code (app.py) and System Prompt Definition

```
> run.ps1 > ...
1 # run.ps1 - Start Ollama + Flask web app with venv auto-activation
2
3 # Path to your virtual environment
4 $venvPath = "..\venv\Scripts\Activate.ps1"
5
6 # Activate venv if it exists
7 if (Test-Path $venvPath) {
8     Write-Host "🟢 Activating virtual environment..."
9     & $venvPath
10 } else {
11     Write-Host "⚠️ Virtual environment not found at $venvPath"
12 }
13
14 # Start Ollama server in background if not already running
15 $ollamaRunning = (Get-Process -Name ollama -ErrorAction SilentlyContinue)
16 if (-not $ollamaRunning) {
17     Write-Host "🚀 Starting Ollama..."
18     Start-Process ollama serve
19     Start-Sleep -Seconds 3 # wait a few seconds for Ollama to bind
20 } else {
21     Write-Host "✅ Ollama already running."
22 }
23
24 # Start the Flask web app
25 Write-Host "🌐 Launching CIMA Web Interface..."
26 python app.py
27
28
```

PowerShell Script (run.ps1) for Automated Environment and Service Startup



HTML Code for the CIMA User Interface and Navigation

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